Reading Financial Conditions in Units of Monetary Policy

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Abstract

How many points of monetary policy is a movement in financial conditions worth? An event study answers by asking how large a conventional policy surprise the movement resembles. A spending calibration asks how large a change in the policy rate would have a comparable effect on GDP. I compare four calibrations of the 10-year yield, equity valuation and the dollar on common series and one scale: market weights from two measures of policy news in FOMC announcement windows, the spending weights of the FRB/US model, and the published Goldman Sachs weights. Knowing how an index was built tells a reader surprisingly little about which other index it will resemble. The futures-based market weights and the published Goldman Sachs weights read a dollar appreciation relative to an equity decline almost identically, while the market weights based on the two-year yield and FRB/US give the dollar more than twice the relative weight. Over 1996–2026 the four readings of annual changes usually give the same directional signal, and materially opposing readings are rare. The size of a tightening is more sensitive. Switching between two implementations of the same method can change the reading as much as switching from market responses to spending effects, and the two closest readings come from different methods. GDP growth provides little evidence for choosing between the futures-based market reading and the FRB/US reading, and the unemployment comparison is sensitive to the sample.

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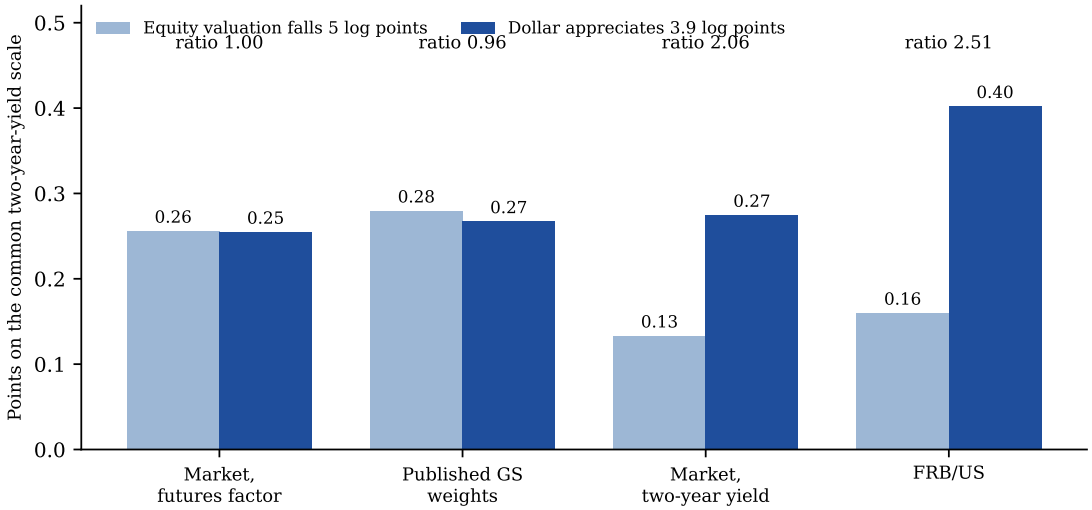


Figure 1: Two Tightenings Read by Four Calibrations

Notes: Readings of each movement under four portfolios of the same three assets relative to r^*, each normalized to read one on the response of the three assets to one unit of policy news measured by the futures factor (points on the common two-year-yield scale). Market: weights from announcement-window responses, with policy news measured by the futures factor or by the change in the two-year yield. FRB/US: effects on GDP after four quarters of persistent shifts. Published GS weights: the U.S. weights on the 10-year yield, equities and the dollar reported by Caballero and Simsek (2026), applied to the series used here with the GS policy-rate and spread terms held fixed. The ratio above each pair is the dollar reading over the equity reading; it does not depend on the normalization. The sizes of the movements were chosen to give equal readings under the futures factor; the ratio of one calibration's ratio to another's does not depend on that choice.

1 Introduction

Suppose the Federal Reserve leaves the policy rate unchanged. In one month stock prices fall; in another the dollar appreciates. Both movements tend to restrain spending. How much tightening has each delivered? One way to answer is to ask how large a monetary policy surprise would ordinarily produce a comparable movement in financial markets. Another is to ask how large a change in the policy rate would have a comparable effect on spending. A policymaker might reasonably expect the two answers to be close.

The two questions are different, and the difference is real. It does not, however, settle the answer. Figure 1 applies four calibrations to the same two movements: a 5 log point fall in equity valuation and a 3.9 log point appreciation of the dollar, with the 10-year yield unchanged. I chose the sizes so that the market index built from the futures-based measure of policy news rates the two alike. Two calibrations read the dollar movement as about as tight as the equity movement: that market index (a ratio of 1.00) and the published Goldman Sachs weights, which are calibrated to effects on GDP (0.96). Two read it as more than twice as tight: the market index built from the change in the two-year yield (2.06) and the Federal Reserve's FRB/US model (2.51). The split runs across the two methods, not between them.

Caballero and Simsek (2026) give a related example with the published Goldman Sachs weights: a 20 percent fall in U.S. equity prices and a 17 percent appreciation of the currency each tighten the index by about one point, because equities carry a small weight and move a lot, and the exchange rate carries a larger weight through expenditure switching. Applied to the movements of Figure 1, the same weights give a ratio of 0.96.

The choice matters for real decisions. In 2008–09, when equity valuations collapsed, all four calibrations registered a large tightening over the year, but the futures-factor and GS readings averaged 2.05 and 2.17 points and the two-year-yield and FRB/US readings 1.33 and 1.40. The direction was clear; the size differed by more than half a point, an amount worth discussing at a policy meeting.

Policymakers read the stance of policy from financial conditions. In November 2023 the Chair noted that "Financial conditions have tightened significantly in recent months, driven by higher longer-term bond yields, among other factors" (Powell, 2023), and in March 2024 that "there are many different financial conditions indicators, and you can kind of, you know, see different answers to that question" (Powell, 2024). A reading of financial conditions is a reading of policy only to the extent that a point of the index means a point of policy. This paper asks how many points of policy a movement in financial conditions is worth under four calibrations, whether knowing how an index was built predicts its reading, whether the choice would have changed the reading of history, and whether later activity favors any calibration.

I find that knowing how an index was constructed tells a reader surprisingly little about which other index it will resemble. I put the four calibrations on one scale, on which each reads one on the futures-factor response to policy news, and apply them to every 12-month change in the three assets from 1996 to 2026. Across the four calibrations, financial conditions usually give the same directional signal. All four readings have the same sign in 81.6 percent of the 369 months, and at least two give materially opposing readings, each at least 25 basis points from zero, in 8 months. The size of the tightening is more sensitive. Switching between the two market calibrations moves the reading by 28 basis points on average, and switching between FRB/US and the published Goldman Sachs weights by 28, against 26 for the futures-factor market index and FRB/US. Switching between two implementations of the same method can change the reading as much as switching from market responses to spending effects. The two closest readings in the sample come from different methods: the futures-factor market index and the published Goldman Sachs weights differ by 7 basis points on average. That is a ranking of point estimates; the distances have wide intervals. The ranking does not come from the transformations of the series: on the real 10-year yield, the log of the cyclically adjusted price-earnings ratio and the real dollar, and on the nominal 10-year yield, the same equity measure and the nominal dollar, the futures-factor and GS readings remain the closest pair, and a within-method distance remains at least as large as the featured cross-method distance.

The paper also explains why the weights differ. On the same announcements, the part of the two-year yield change that the futures factor does not capture moves long yields and the dollar and not equities, so a market index built to recover the two-year yield puts less weight on equities. In

FRB/US, net exports carry almost all of the effect of the dollar on GDP, and consumption most of the effect of equities. Why the published Goldman Sachs weights line up with the futures-factor market weights is a finding whose cause the paper leaves open.

The exceptions to directional agreement are concentrated. At 25 basis points, adding the other two calibrations produces no month of material opposition beyond the 8 of the futures-factor and FRB/US pair; at 10 basis points it doubles the count, from 19 to 38. In 6 of the 8 months of materially opposing readings, FRB/US alone reads a tightening that the others read as an easing, in late 1997 and 1998 and in early 2015, when the dollar and valuations had both risen by more than 10 percent over the year. In the other 2, in 2003 and 2008, the futures-factor and Goldman Sachs readings were tight and FRB/US was easy. The four readings are furthest apart in 2008 and 2009, and in 2022–24 every calibration shows a tightening of about one point or more. Aggregate unemployment and GDP growth do not separate the calibrations, and the tests have little power.

The paper has five further parts. Section 2 defines a policy equivalent under each method. Section 3 explains why the weights differ: the measure of policy news on the market side, the expenditure channels and the published weights on the spending side, and the role of persistence in the conversion. Section 4 puts the four calibrations on one scale, reads three episodes with them and asks how often the choice changes the judgment over the full history. Section 5 asks what subsequent activity can establish, and Section 6 examines the stability of the weights and the limits of the design.

1.1 Connection to the Literature

Weights and their meaning. Indexes of financial conditions differ mainly in the source of their weights (Hatzius et al., 2010): effects on output from a model (Dudley and Hatzius, 2000; Hatzius and Stehn, 2018; Ajello et al., 2023), principal components (Brave and Butters, 2011), or event-study regressions (Kuttner, 2001; Gürkaynak, Sack and Swanson, 2005; Bernanke and Kuttner, 2005; Nakamura and Steinsson, 2018; Bauer and Swanson, 2023). The older literature on monetary conditions indexes recognized that the weights on asset prices depend on the shock and the dynamics, so that their policy meaning is not invariant (Eika, Ericsson and Nymoen, 1996). I quantify that difficulty with four sets of U.S. weights on common assets and ask where it matters.

The closest comparisons. Arnaut and Bauer (2024) estimate how an existing index responds to policy surprises, Sack and Swanson (2023) relate FCI-G to policy-induced changes in the two-year yield, and Ajello et al. (2023) translate movements into a growth impulse with model multipliers. Bletzinger, Martorana and Mistak (2026) build a U.S. macro-finance index in units comparable to a policy rate and compare it with the Goldman Sachs index and FCI-G, finding divergences driven by equities and the strongest tightening in 2022–24. Comparisons of published indexes mix the weights with differences in assets, transformations and units. My design holds the assets, the transformations and the policy unit fixed across calibrations, so that a difference in readings can be attributed to the weights. Boehm and Kroner (2025) show that announcements also move equities and exchange rates along a dimension orthogonal to yield changes, so a movement that departs

from the conventional pattern can come from the announcement itself.

Information about activity. Hatzius et al. (2010) build an index conditional on past activity and inflation, Aramonte, Rosen and Schindler (2013) show that comparisons among indexes depend on the crisis in the sample, and Lombardi, Manea and Schrimpf (2025) separate safe yields from priced risk and find that risk predicts activity more strongly. Adrian, Boyarchenko and Giannone (2019) show that conditions shift the lower quantiles of growth, and Brave and Butters (2011) build an index of risk, credit and leverage. Caballero, Caravello and Simsek (2026) define FCI*, the level of financial conditions consistent with output at potential. The neutral rate matters for the level of every index (Laubach and Williams, 2003; Holston, Laubach and Williams, 2017, 2023; Hamilton et al., 2016; Bauer and Rudebusch, 2020; Kiley, 2020); I measure the inputs relative to r^* and do not estimate a neutral level for the index (Appendix B). **Why the meaning of a point matters.** Caballero, Caravello and Simsek (2024) study targeting an index of financial conditions in a closed economy, and Caballero and Simsek (2026) extend the analysis to a multi-asset open economy, in which domestic asset prices and the exchange rate both enter aggregate demand and policy responses link the markets. That gives a reason to care which mix of asset prices delivers a given index reading. My question comes one step earlier: when does a portfolio inferred from conventional policy news give those markets the same relative importance as a spending calibration? The answer does not require endorsing targeting. The same ambiguity arises for anyone who communicates about a desired path of financial conditions, since the path has to say what a point on it means, and for a neutral benchmark such as FCI* (Caballero, Caravello and Simsek, 2026), which takes the aggregation of asset movements as given.

2 Two Ways to Read a Financial Movement

2.1 Two Meanings of a Policy Equivalent

Take a movement in three asset prices: the 10-year Treasury yield, equity valuation and the real broad dollar. Each is measured relative to r^* and to its 1973–2026 average (Appendix B). I call a reading of the movement a policy equivalent if it says how much tightening of the policy rate the movement amounts to. There are two natural ways to compute one.

The market index starts from policy. At each Federal Open Market Committee announcement I measure the policy surprise with the factor of Nakamura and Steinsson (2018), scaled so that one unit raises the two-year yield by one percentage point, and I measure how the three asset prices respond in a narrow window around the announcement. The weights are chosen to recover the size of the surprise from the three asset prices together. A movement is then worth the size of the surprise that would, on average, produce it, in points of the two-year yield. A policy surprise of one unit reads one. I call this the market-implied reading.

The FRB/US index starts from spending. Each asset price is weighted by its effect on GDP four quarters later in the Federal Reserve's FRB/US model (Brayton, Laubach and Reifschneider, 2014), in the experiment in which the asset price shifts persistently, divided by the effect of a persistent

one percentage point rise in the federal funds rate. A movement is then worth the funds-rate change with the same effect on GDP, in funds-rate points. I call this the spending-equivalent reading.

The first calculation reads a movement through what policy does to markets. The second reads it through what markets do to spending. Both are legitimate. They need not agree.

2.2 Why the Weights Differ

Consider the dollar. It can be a reliable indicator of a policy surprise because it responds consistently when the expected path of U.S. interest rates changes. How useful it is as an indicator depends on how much it also moves for other reasons, and on what yields and equities already reveal. Its weight in a spending index depends on different things: how an appreciation affects net exports, and how long the appreciation lasts. An asset can be an excellent indicator of a policy surprise and have a modest effect on spending. It can also be a noisy indicator and have a large effect on spending. The same is true of equities, whose effect on spending runs through wealth and the cost of capital. A fall in equity prices that reflects a higher discount rate is policy-like, and a fall that reflects weaker earnings prospects is not policy at all. The event study averages over announcements, where the first kind dominates. The model prices an equity movement by its estimated effect on spending, whatever its source. Neither calculation is wrong about its own question.

Two equations make the argument precise. Let x be the movement in the three asset prices around an announcement, s the policy surprise, and ψ the vector of responses of the asset prices to a unit of surprise:

$$x = \psi s + u. \tag{1}$$

The vector u collects the ways in which a particular movement departs from the usual response to conventional rate news. It has covariance Σ. Even an announcement can produce such a departure. The market weights w_M recover s from x by generalized least squares, $w_M \propto \Sigma^{-1}\psi$, normalized so that $w_M'\psi = 1$. The FRB/US weights b_h are the effects of the three asset prices on GDP at horizon h in funds-rate points. The model prices the usual response ψ at $b_h'\psi$, which I report in its own units, and dividing by it gives the normalized weights $w_{B,h} = b_h/(b_h'\psi)$, so that both indexes read one on a conventional policy surprise. The difference between the two readings of a movement x is then

$$D_h(x) = (w_{B,h} - w_M)'x = (w_{B,h} - w_M)'u, \tag{2}$$

because the policy part ψs cancels. The same difference has a second reading. The fitted policy component of any movement x is ψM, where $M = w_M'x$ is the market-implied reading; the remainder $x - \psi M$ is the combination of asset prices that a conventional policy surprise cannot reproduce. Once the scales are matched, the spending-equivalent valuation of that remainder is precisely the difference between the two readings,

$$D_h(x) = w_{B,h}'(x - \psi M), \tag{3}$$

because $w'_{B,h}\psi = 1$. A reader can say what the difference measures without assigning it a structural cause. The argument extends to any two portfolios normalized to read one on the usual policy response: their difference prices only the part of an observed movement outside that response. That is the bridge from two methods to four calibrations. It also means that an arbitrary movement can read as one point of tightening under every calibration, and so resemble a policy tightening, even when no policy shock produced it. In words, the two methods disagree only to the extent that asset prices move in unusual proportions and the methods value those departures differently. An unusual movement is not enough by itself: if the weights are the same, there is nothing to value differently. A difference in weights is not enough either: if the asset prices keep moving together as they do around policy news, the difference has nothing to act on. That is why two indexes can track each other closely for decades and still give different readings during an equity boom or a dollar surge. Throughout, D is the difference computed on the movements of the three asset prices over the horizon in question, and in Section 5 it is computed on 12-month changes. Appendix A derives the weights.

2.3 Putting Numbers on It

Figure 1 applies four sets of weights to two movements. Neither movement is a policy surprise, so nothing in the construction requires the readings to agree. For an equity decline of size e and a dollar appreciation of size d, the ratio of one calibration's dollar-to-equity reading to another's does not depend on e or d, because d/e cancels; the choice of movements fixes the futures-factor ratio at one and sets the gap in basis points, not the relative weights. Table C.8 gives the FRB/US readings in its own units, funds-rate points, before any normalization. Whether the differences are large in practice depends on how often asset prices move in such proportions, which Sections 4.1 and 4 measure.

3 Where the Weights Come From

Table 1 puts the four sets of weights side by side on the common scale, on which each reads one on the response of the three assets to one unit of futures-factor policy news. The futures-factor market weights and the published GS weights are close on all three assets. The two-year-yield market weights put more weight on the 10-year yield and half as much on equities. FRB/US puts the most weight on the dollar relative to equities. This section explains the two differences the paper can trace: why a different measure of policy news lowers the market equity weight, and why the dollar matters for GDP in FRB/US.

3.1 One Set of Announcements, Two Measures of Policy News

The market weights come from the 253 announcements from January 1996 to September 2026 in the San Francisco Fed's U.S. Monetary Policy Event-Study Database (Acosta et al., 2025). In a narrow window around each announcement I measure the responses of the 10-year yield, the S&P 500 and

Table 1: Four Sets of Weights on One Scale

	Market, futures factor	Published GS weights	Market, two-year yield	FRB/US
Weight, 10-year yield (per point)	0.609	0.515	0.926	0.590
Weight, equity valuation (per log point)	−0.051	−0.056	−0.027	−0.032
Weight, dollar (per log point)	0.065	0.069	0.070	0.103
Equity valuation falls 5 log points	0.26	0.28	0.13	0.16
Dollar appreciates 3.9 log points	0.25	0.27	0.27	0.40
Dollar reading over equity reading [90%]	1.00 [0.33, 2.25]	0.96	2.06 [0.82, 5.19]	2.51

Notes: Weights on the three assets relative to r^*, each set normalized to read one on the response of the three assets to one unit of futures-factor policy news; readings of the movements of Figure 1 in points on that common scale. Market weights: one-factor generalized least squares on 253 announcement windows, with policy news measured by the futures factor or by the change in the two-year yield. FRB/US: effects on GDP after four quarters of persistent shifts, divided by the effect of a persistent one-point rise in the funds rate, then normalized. Published GS weights: the U.S. weights on the 10-year yield, equities and the dollar, with the GS policy-rate and spread terms held fixed, then normalized. Intervals over 5,000 bootstrap draws of the announcements, for the estimated weights.

Table 2: Market Index: Responses to Policy News and Weights

	Response	Weight	90% interval	Share of policy move
10-year Treasury yield (pp)	0.53 (0.06)	0.609	[0.147, 1.114]	0.33
Equity valuation, 100 ln CAPE (percent)	−7.78 (0.93)	−0.051	[−0.068, −0.030]	0.40
Real broad dollar, 100 ln (percent)	4.22 (0.53)	0.065	[0.021, 0.112]	0.28

Notes: Announcement windows, 253 announcements, January 1996 to September 2026. Response: change in the window per unit of policy news, which moves the two-year yield by 1 percentage point (heteroskedasticity-robust standard errors in parentheses). Weight: one-factor generalized least squares, normalized so that the weights applied to the responses sum to 1. Interval: 5th and 95th percentiles of weights re-estimated on 5000 bootstrap samples of the announcements. Share: weight times response.

the dollar. The weights recover a specified measure of the policy news from the three responses together, so the definition of policy news is a measurement choice. Two measures are natural on these announcements. The baseline is the futures-based factor of Nakamura and Steinsson (2018). The alternative is the change in the two-year Treasury yield in the same windows, which is the unit in which the market index is quoted.

The two measures share most of their variation. Their correlation is 0.90. Write the two-year change as the factor plus a remainder. The remainder is 20 percent of the variance of the two-year change (interval 12 to 30 percent), about 44 percent of its standard deviation. Figure 2 shows which assets move with each part. A point of the factor raises the 10-year yield by 0.53 points, lowers the S&P 500 by 7.78 percent and raises the dollar by 4.22 percent. A point of the remainder is associated with a rise in the 10-year yield of 0.80 points and in the dollar of 4.30 percent, and with an equity response of 0.55 percent (−5.14 to 5.23) that the data cannot distinguish from zero.

Table 2 gives the market weights under the baseline. A policy move that raises the two-year yield by one point raises the 10-year yield by 0.53 points, lowers equity valuation by 7.78 percent and raises the real dollar by 4.22 percent. Each variable carries a substantial part of the policy move, with shares of 0.33 for the 10-year yield, 0.40 for equity valuation and 0.28 for the dollar.

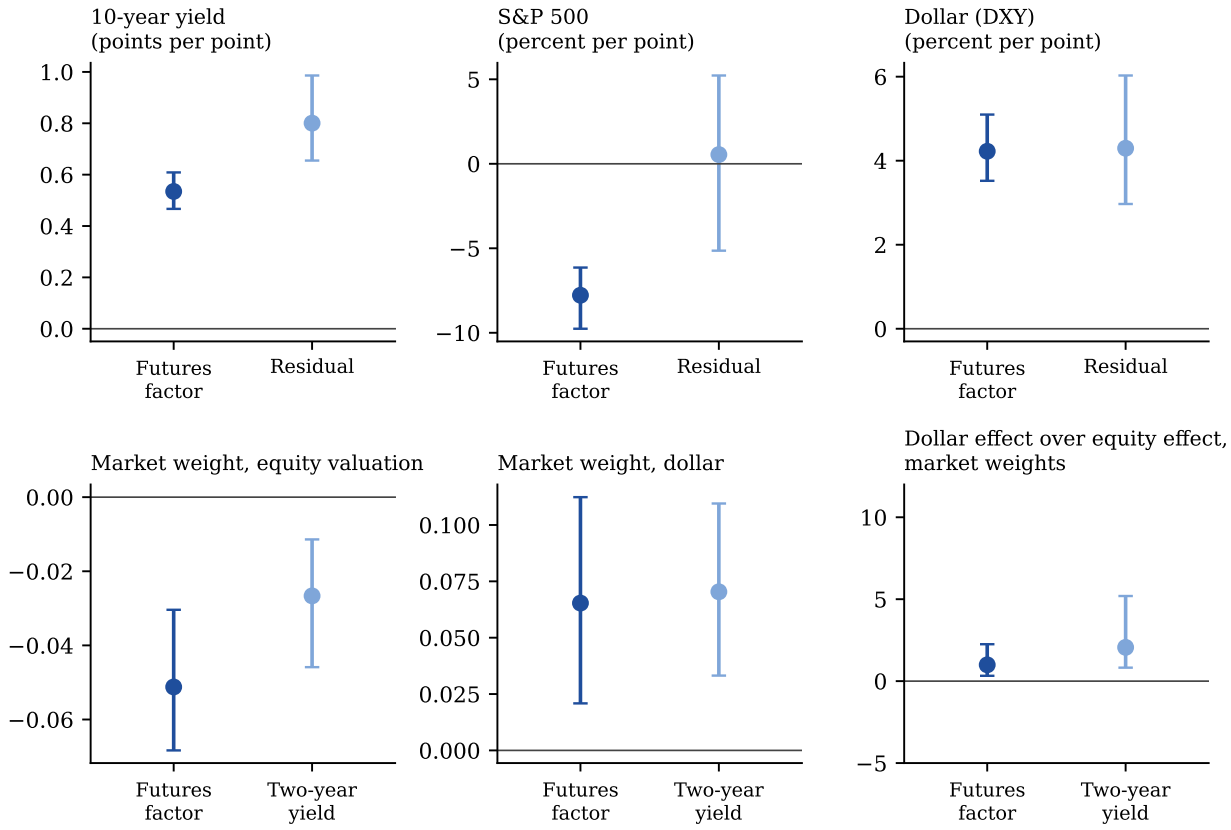


Figure 2: What the Two Measures of Policy News Capture

Notes: Top: responses of the 10-year yield (points), the S&P 500 and the dollar (DXY) (percent) in the announcement windows to one point of the futures factor and to one point of the part of the two-year yield change that is orthogonal to it; the factor is scaled to the two-year change. Bottom: the market weight on equity valuation and on the dollar, and the market's dollar-over-equity effect ratio, when policy news is measured by the futures factor and by the two-year yield. Bars are 90 percent intervals from 5,000 bootstrap draws of the announcements, drawn in common for the two signals.

Why the equity weight falls. The weights are the generalized least squares weights that recover the signal, $w_M \propto \Sigma^{-1}\psi$. When the signal is the two-year yield, the remainder becomes part of the signal and not part of the noise. Because the remainder moves yields and the dollar and not equities, equity valuation responds less per unit of the signal and is a weaker indicator of it. The equity weight falls from -0.051 to -0.027. Switching the response vector and the residual covariance in turn, each accounts for about half of the change (0.011 and 0.013), and the order does not matter. The market's dollar-to-equity weight ratio under the two-year yield is 2.07 times its ratio under the factor, with a paired 90 percent interval of 0.98 to 5.47. I report the roughly twofold change as a point estimate. The regressions show what variation each measure captures. They do not show whether the remainder is further policy news, a term-premium movement or measurement error.

What the weights say. The weights say what the variables do when policy moves, not what causes their movements at other times. A rise in stock prices that comes from news about earnings has little to do with policy, and the index treats it as if it did. An announcement can also reveal the Fed's view of the economy, and then the responses mix policy and information (Nakamura and Steinsson, 2018). I measure the responses as markets receive them and set the information effect aside. The index measures policy-equivalent tightening. It is not a forecast of activity.

Readings of the two movements. Table C.7 in the appendix applies the weights, normalized for each signal, to the movements of Figure 1. The market-implied reading rates the equity decline and the dollar appreciation 0.00 apart with the futures factor and 0.14 apart with the two-year yield. The model rates the dollar movement 0.24 points tighter than the equity movement. The model less market contrast is 0.243 (90 percent interval 0.003 to 0.448) for the baseline and 0.107 (-0.079 to 0.287) for the two-year yield. On common draws, switching from the futures factor to the two-year yield lowers the contrast by 14 basis points of the two-year yield (-0.14, interval -0.27 to 0.03). The model gives the dollar more weight relative to equities than the market does in the point estimates under both measures, and the interval for the difference includes zero under the two-year yield.

Proportionality. Whether the model's dollar-to-equity weight ratio equals the market's is a question about relative weights and not about size. The cross-product $c = w_{B,\$}w_{M,E} - w_{B,E}w_{M,\$}$ is zero if and only if they are equal. Its 90 percent interval excludes zero for the futures factor (-0.0032, -0.0058 to -0.0004) and includes it for the two-year yield (-0.0005, -0.0029 to 0.0015). The ratio of the model's ratio to the market's, which is 2.53 for the baseline and 1.22 for the two-year yield (interval 0.48 to 3.05), tests the same hypothesis on a scale-free ratio, but its denominator, the market's equity effect, is small and imprecise. The two sample checks, announcements through 2019 and outside the lower-bound years, give contrasts of 0.349 (0.036 to 0.565) and 0.232 (0.021 to 0.405); Appendix D.1 reports them. The intervals are not adjusted for the number of definitions.

3.2 How Precisely Are the Weights Estimated?

The weights come from regressions on 253 announcements. The weight on the 10-year yield has a 90 percent interval from 0.15 to 1.11 around 0.61. The interval for equity valuation runs from -0.07 to -0.03, and for the dollar from 0.02 to 0.11. The FRB/US weights on all three lie inside these intervals, the equity weight near the edge: it is lower in absolute value than 5 percent of the bootstrap draws, about 14 percent in a two-sided test. The data do not reject that the model weights equal the market weights. This is a statement about power, not a finding that they are equal, and it treats the FRB/US weights as fixed. The intervals in the text come from the bootstrap of the announcements; the contrast in composition in Section C.4 is the ratio of ratios, whose interval includes one at two quarters.

The intervals behind Figure 1 and the FRB/US readings in funds-rate points are in Table C.8 in the appendix.

3.3 Where the Model's Weights Come From

Table C.9 compares the market weights with the FRB/US weights. The two sets agree on the 10-year yield, where the FRB/US weight is 0.92 times the market weight. The model puts less weight on equity valuation (0.60 times) and more on the dollar (1.51 times). On a policy move the difference does not matter, because the policy part cancels in equation (2). On a departure from the usual response, the model gives a fall in equity prices its estimated effect on output, and the market index treats it as policy-equivalent.

The FRB/US weights are effects on GDP, and a reader should see where they come from. Figure 3 follows the two movements of Figure 1 through the model under one persistence assumption, a persistent shift. Only the stated variable moves. The policy rate, the Treasury curve, the mortgage and car-loan rates, the BBB spread, equity valuation, household stock wealth and the dollar are held at baseline in every experiment except their own, so the policy rule is inactive and expectations are those of the model's VAR. The GDP object is the percent deviation of real GDP from baseline. The contributions are the model's own expenditure identities, weighted by shares in GDP, and they sum to GDP.

The equity movement reduces GDP by -0.070 percent after four quarters, and domestic demand accounts for it. Consumption supplies 92 percent of the effect, through wealth and the cost of capital, and business fixed investment 27 percent. Domestic demand in all supplies 120 percent, and net exports offset 33 percent of it, because weaker demand reduces imports; inventories supply 13 percent and government -1 percent, so that the shares sum to 100. The dollar movement reduces GDP by -0.177 percent, and net exports account for 94 percent of it: exports 73 percent and imports 21 percent. Domestic demand contributes 0.6 percent, inventories 13 percent and government -7 percent of the GDP effect. With a half-life of four quarters the shares are similar (consumption 91 percent of the equity effect and net exports 91 percent of the dollar effect). The two weights therefore reflect two different channels, and the dollar's larger weight is a statement

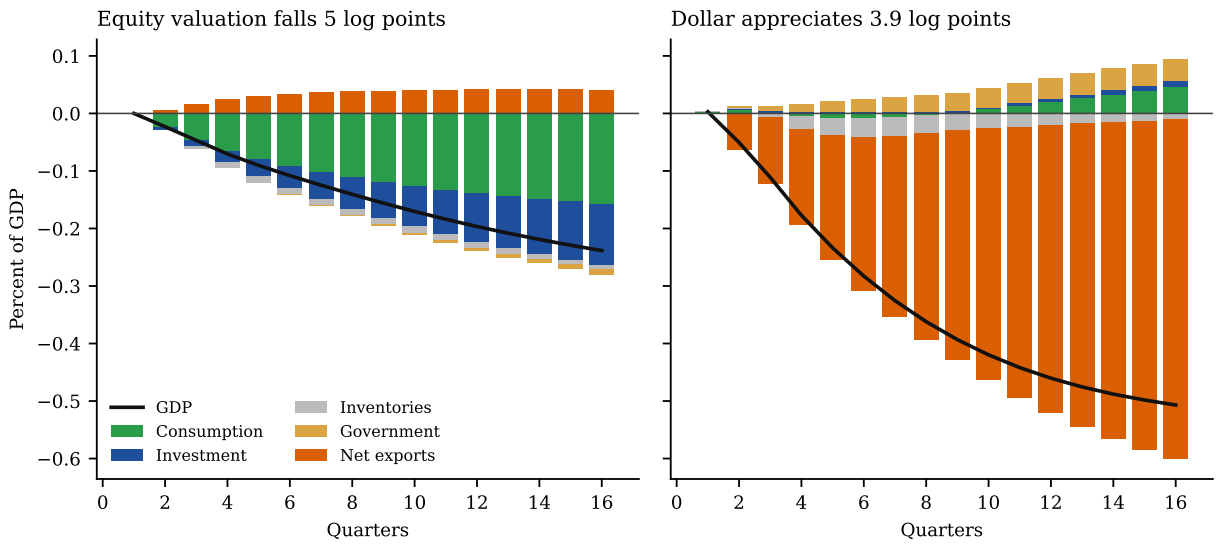


Figure 3: How the Two Movements Reach GDP in FRB/US

Notes: Percent deviation of real GDP from baseline and contributions by expenditure component, persistent shift in the variable, other financial variables held at baseline, policy rule inactive. Investment is residential and business fixed investment; net exports are exports less imports. The figure uses the Board's FRB/US package with the data vintage of October 2026, the vintage behind every FRB/US number in the paper.

about trade. It carries no implication about how persistent the dollar is, which is the assumption of the experiment.

The experiments need two mappings from the market series to the model. The equity experiment shocks the required return on equity, and household stock wealth falls with it in the same simulation, so valuation and wealth move jointly; the 5 log point fall in valuation is entered as a 5 percent fall in stock wealth on impact. The market side measures equity by the S&P 500 return in the announcement window, which gives the weight on equity valuation, and applies that weight to monthly changes in the earnings-yield gap converted to CAPE log points (Appendix B); a one percent window return is treated as one log point of valuation, with earnings unchanged. The dollar experiment raises the model's real exchange rate by one percent per unit, and the market side measures the dollar by the DXY in the window and applies the weight to the log of the broad real dollar. DXY is a nominal index of six currencies and the model's dollar is a trade-weighted real index. I treat them as the same unit, and over a window or a month the nominal moves dominate.

The model's own policy shock gives a funds-rate equivalent of 1.77 for the usual announcement mix against 0.95 for the announcement response. FRB/US can therefore assign a value near one to the observed announcement mix even though its own policy shock generates a different mix: heavier in equities and lighter in the dollar. The comparison is not a decomposition of how much of transmission the three assets omit, because the policy shock changes the path, the equity response, the channels that operate and the unit of normalization together. Appendix D.2 gives the

experiment.

A check on the spending side. The multipliers of the Board's FCI-G offer a check, with the caveat that they share a lineage with FRB/US, so agreement measures robustness within a family. Per 1 percent change in each variable, the FCI-G dollar multiplier is 2.25 times its equity multiplier, against 3.22 in FRB/US (these ratios use the multipliers per 1 percent, not the sizes of the movements of Figure 1, for which the FRB/US ratio is 2.51). The ratio of ratios against the market is 1.76 under the baseline signal and 0.85 under the two-year yield. FCI-G corroborates the direction under the futures factors and not under the two-year yield. A genuinely independent check would use another model with documented asset-specific multipliers.

3.4 The Published Goldman Sachs Weights

The fourth calibration uses the published weights of the Goldman Sachs U.S. financial conditions index. As Caballero and Simsek (2026) describe them, the weights are calibrated to each variable's estimated effect on real GDP growth one year ahead in a macroeconomic model, and a one-point rise in the index is estimated to lower GDP by about 1 percent after four quarters. The U.S. weights are 4.4 on the policy rate, 45.1 on the 10-year yield, 39.6 on a corporate spread, 4.9 on an equity variable (prices scaled by a 10-year average of earnings) and 6.0 on a trade-weighted dollar. I apply the weights on the 10-year yield, equities and the dollar to the series used for the other three portfolios, relative to r^*, and hold the policy-rate and spread terms fixed, as the FRB/US index's other variables are held fixed. The resulting portfolio is the published GS weights on the common assets, not the GS index. Normalized to the common scale, its weights are 0.515 on the 10-year yield, -0.056 on equity valuation and 0.069 on the dollar, against 0.609, -0.051 and 0.065 for the futures-factor market index. Like FRB/US, the published weights measure spending effects, and they share part of its modeling lineage; unlike FRB/US, they give equities and the dollar nearly the relative weights that recover futures-based policy news. The close match is a finding. Its economic cause is open.

3.5 The FRB/US Reading of a Policy Move

Under persistent asset paths, FRB/US assigns 0.954 funds-rate points (interval 0.86 to 1.08) to the asset response associated with a surprise that raises the two-year yield one point. With a four-quarter half-life it assigns 0.82, and at two and eight quarters with persistent paths 0.901 and 0.917 (Table C.10). I use the conversion under persistent paths at four quarters to compare how the methods weight individual assets. After division by it, the FRB/US readings are in points on the common two-year-yield scale, the scale of Figure 1 and of every comparison below. The empirical finding is the sensitivity of the conversion to the assumed path. Persistence changes the conversion and not the relative weights: with the same decay for both assets, the model's dollar-to-equity ratio is 2.53, 2.55 and 2.59 at half-lives of 8, 4 and 2 quarters (Appendix C.4). Exact agreement along the policy direction after the division follows from the units and is not evidence. The model's own policy shock answers a further question and is in Appendix D.2.

Table 3: How Far Apart Are the Four Readings?

	Market, futures factor	Published GS weights	Market, two-year yield	FRB/US
Market, futures factor	–	7 (0.09)	28 (0.37)	26 (0.33)
Published GS weights	7 (0.09)	–	34 (0.45)	28 (0.36)
Market, two-year yield	28 (0.37)	34 (0.45)	–	23 (0.30)
FRB/US	26 (0.33)	28 (0.36)	23 (0.30)	–

Notes: Mean absolute difference between two readings of the same 12-month changes in the three assets relative to r^*, monthly, January 1996 to September 2026 (369 months), in basis points of the two-year yield; in parentheses, the same divided by the mean absolute reading of the market index with the futures factor (the common denominator). Every portfolio is normalized to read one on the futures-factor response to policy news, so that the distances reflect relative weights. Published GS weights: the U.S. weights on the 10-year yield, equities and the dollar, applied to the same series, with the GS policy-rate and spread terms held fixed. Intervals over 5,000 bootstrap draws of the announcements are in the text.

4 What Changes in History

4.1 Four Readings on One Scale

Each of the four portfolios is a set of weights on the same three assets. I normalize each to read one on the same vector, the response of the three assets to one unit of futures-factor policy news. Agreement along that vector follows by construction. The evidence comes from how the portfolios read the other combinations of asset movements that occurred. The normalization barely changes the two-year-yield market weights, which already read 0.999 on that vector because the two measures of policy news share most of their variation. It matters for FRB/US and the published GS weights, whose raw units are funds-rate points and GS index points.

I apply the four portfolios to the 12-month change in the three assets in every month from 1996 to 2026, 369 overlapping months. Table 3 gives the mean absolute difference between every pair of readings, in basis points of the two-year yield and as a share of the mean absolute reading of the futures-factor market index.

The closest pair crosses the methods. The futures-factor market index and the published GS weights differ by 7 basis points on average, 0.09 of the typical market move. The two pairs within a method are among the widest: 28 basis points between the two market calibrations and 28 between FRB/US and the published GS weights, against 26 between the futures-factor market index and FRB/US and 23 between the two-year-yield market index and FRB/US. Switching between two implementations of the same method can therefore change the reading as much as switching from market responses to spending effects.

These are rankings of point estimates. Over 5,000 bootstrap draws of the announcements, the futures-factor to GS distance has a 90 percent interval of 0.07 to 0.57 of the typical move. A distance cannot be negative, so when two sets of weights nearly coincide, sampling error in one of them tends to raise the distance; the point estimate sits near the bottom of its interval for that reason. The interval for the FRB/US to GS distance, 0.30 to 0.44, is narrow because neither set of weights is estimated here; only the common normalization varies. The paired difference between

Table 4: The Four Calibrations on Other Input Series

	Relative to r^* (paper)	Real, 100 ln CAPE	Nominal, 100 ln CAPE, nominal dollar
Mean absolute distance (basis points)			
Market, futures factor vs Published GS weights	7	8	10
Market, futures factor vs FRB/US	26	25	26
Market, two-year yield vs FRB/US	23	25	32
Market, futures factor vs Market, two-year yield	28	35	39
FRB/US vs Published GS weights	28	31	32
Market, two-year yield vs Published GS weights	34	43	49
Judgments, 369 months			
All four the same sign (percent)	81.6	72.6	73.4
Materially opposing months, 25 bp	8	16	26
Materially opposing months, 10 bp	38	63	62
2008–09, average reading (points)			
Market, futures factor	2.05	1.00	0.88
Published GS weights	2.17	1.15	1.06
Market, two-year yield	1.33	0.31	0.10
FRB/US	1.40	0.58	0.50

Notes: Each column applies all four sets of weights, unchanged and normalized to the futures-factor response, to the same series: the paper's series relative to r^*; the real 10-year yield, equity valuation as 100 times the log of the cyclically adjusted price-earnings ratio (prices over a 10-year average of earnings, the documented GS equity concept) and the real dollar; and the nominal 10-year yield, the same equity valuation and the nominal broad dollar, the nearest publicly reproducible GS inputs. Twelve-month changes, January 1996 to September 2026.

the futures-factor to FRB/US distance and the distance between the two market calibrations is −3 basis points (−19 to 27): the two are comparable in this sample, and the data cannot rank them.

Other input series. The four portfolios use the same transformed series, so a reader should ask whether the transformations produce the closeness. Table 4 applies all four sets of weights, unchanged and with the same normalization, to two other sets of series: the real 10-year yield, equity valuation measured as 100 times the log of the cyclically adjusted price-earnings ratio (the documented GS equity concept) and the real dollar; and the nominal 10-year yield, the same equity measure and the nominal broad dollar, the nearest publicly reproducible GS inputs. The futures-factor and GS readings remain the closest pair, 8 and 10 basis points apart against 7 on the paper's series, and the distance between the two market calibrations (35 and 39) and between FRB/US and GS (31 and 32) remains at least as large as the futures-factor to FRB/US distance (25 and 26). The transformations matter for the judgments themselves. On the level series all four readings share a sign in 73.4 percent of months, against 81.6 on the paper's series, and the 2008–09 tightening is much smaller under every calibration, because measuring yields and valuations relative to r^* adds the fall in r^* to the tightening. On common draws, the futures-factor to GS distance less the nearest other distance has an interval that includes zero under every set of series, so the ranking of the closest pair is a point-estimate ranking.

4.2 Three Episodes

Table 5 reads three episodes with the four calibrations, and Figure 4 follows the four readings month by month. All readings are retrospective estimates, made with weights estimated on the

Table 5: Four Readings of Three Episodes

Episode (months)	Market, futures factor	Published GS weights	Market, two-year yield	FRB/US	Mean range	Opposing months
2008–09 (January 2008 to December 2009)	2.05	2.17	1.33	1.40	1.06	1
1998–2001 (January 1998 to December 2001)	−0.06	−0.03	−0.07	0.16	0.49	3
2022–24 (March 2022 to December 2024)	1.02	0.97	1.21	1.01	0.32	0

Notes: Average over the months of each episode of the 12-month change in financial conditions under each portfolio, in points on the common two-year-yield scale (normalized to the futures-factor response). Mean range: average over the months of the largest less the smallest reading. Opposing months: months with at least one reading at least 0.25 above zero and one at least 0.25 below. The readings use the 10-year yield, equity valuation and the dollar only; corporate spreads, which the published GS index and the full FRB/US index contain, are held fixed.

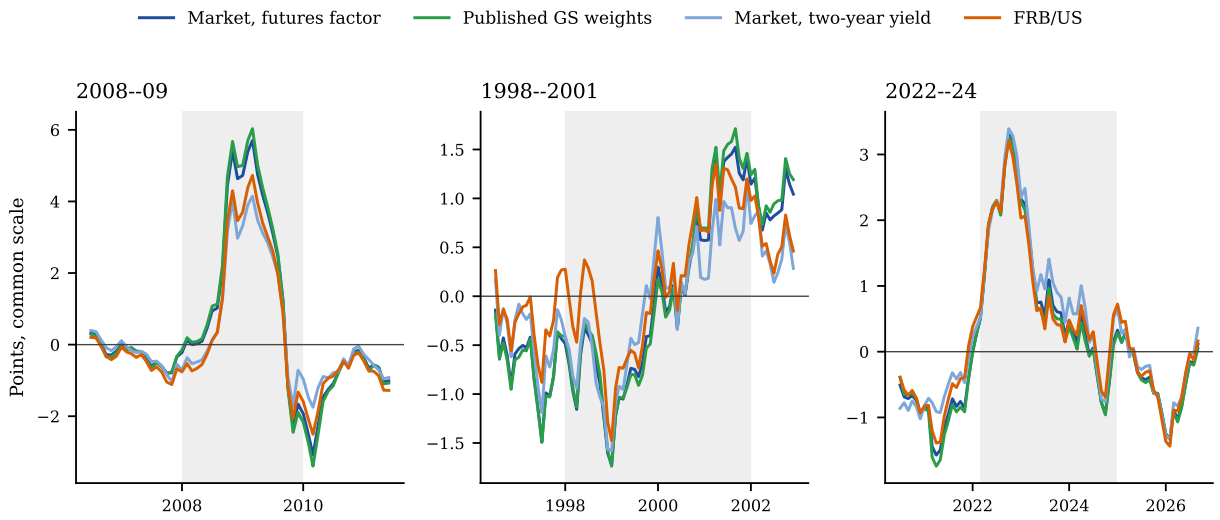


Figure 4: Four Readings, Month by Month

Notes: Twelve-month change in financial conditions under each portfolio, in points on the common two-year-yield scale, normalized to the futures-factor response. The shaded months are the episodes of Table 5. Corporate spreads are held fixed.

full sample, and they use the 10-year yield, equity valuation and the dollar only; corporate spreads, which widened sharply in 2008, are held fixed.

2008–09: the direction is clear, the size is not. From January 2008 to December 2009 the futures-factor and GS readings average 2.05 and 2.17 points of tightening and the two-year-yield and FRB/US readings 1.33 and 1.40. Equity valuations fell by about half, and the futures-factor and GS portfolios give equities the most weight relative to the dollar and the 10-year yield, so the collapse weighs more in their readings. The four readings are furthest apart in 2009 and 2008, with mean ranges of 1.10 and 1.03 points. A policymaker could describe the direction with confidence; the choice of weights changed the estimated tightening by more than half a point. These are readings of three asset prices. Corporate spreads would add to every reading in 2008, by an amount that depends on the spread weight each calibration assigns.

1998–2001. The four readings average −0.06 (futures factor), −0.03 (GS), −0.07 (two-year

yield) and 0.16 (FRB/US): three slightly easing and FRB/US slightly tightening, with materially opposing readings in 3 months. For the pair of the futures-factor market index and FRB/US, averaged over the episode, the market-implied reading of the 12-month changes was slightly easing, at −0.06 points (interval −0.37 to 0.27), and the spending-equivalent reading was slightly tightening, at 0.16 (0.14 to 0.17). The difference of 0.22 (−0.12 to 0.53) is a point estimate; its interval includes zero, and so do the intervals for the separate contributions of the dollar (−0.05 to 0.36) and of equities (0.00 to 0.10). The signs differ in the averages and in 7 of the 48 months, and the monthly paths show what the average conceals (Figure 5). The difference was largest in the 12-month changes ending in the first half of 1998, at 0.76 points in January 1998, because those changes include the dollar's appreciation and the rise in valuations of 1997, and the model gives the dollar more weight and equities less. It shrank through 1999 and 2000 and turned negative in the second half of 2001 as valuations fell. The disagreement was therefore mostly about size and not direction: the signs differ in only 7 of 48 months, but the readings differ by at least a quarter point in 21. The dollar appreciated by 4.18 log points a year on average, which adds 0.16 to D; equity valuations rose by 2.70 log points a year, which adds 0.05. Unemployment rose by 0.35 points over the following year, against forecasts of 0.67 from the market-implied reading and 0.78 from the spending-equivalent reading.

2022–24. Every calibration shows a large tightening: the four readings average 1.02, 0.97, 1.21 and 1.01, and no month has materially opposing readings. For the futures-factor pair, the two readings of the 12-month change averaged 1.02 points (interval 0.69 to 1.35) and 1.01 (0.90 to 1.13). Both movements tightened conditions. Equity valuations fell by 5.04 log points a year, and the dollar appreciated by 2.85. Because the model gives equities less weight than the event study, the fall in equities makes the spending-equivalent reading less tight than the market-implied one, and contributes −0.10 to D. Because the model gives the dollar more weight, the appreciation makes the spending-equivalent reading tighter, and contributes 0.11. The two contributions to the disagreement cancel, so that D averages −0.01 (−0.27 to 0.26), although each asset tightened conditions. The monthly paths show that the cancellation is not exact at every date: D ranged from about −0.3 to +0.5 within the episode, and it exceeded 0.25 in absolute value in 6 of 34 months. The baseline cancellation is specific to that definition of policy news: with the two-year yield as the signal, D is −0.17 and the market-implied reading is higher, and without the lower-bound years D is −0.19 (Section 3.1). A policymaker would have described a sizable tightening under every one of them. The full indexes differ in average level by 0.96 points (interval 0.00 to 1.76), with the equity weights (0.54) and the dollar (0.52) both contributing, and that difference does not affect the readings of the year's tightening. Unemployment rose by 0.27 points over the following year, against forecasts of 0.65 from the market-implied reading and 0.66 from the spending-equivalent reading. Both weightings overpredicted the rise, so the weights do not explain the miss.

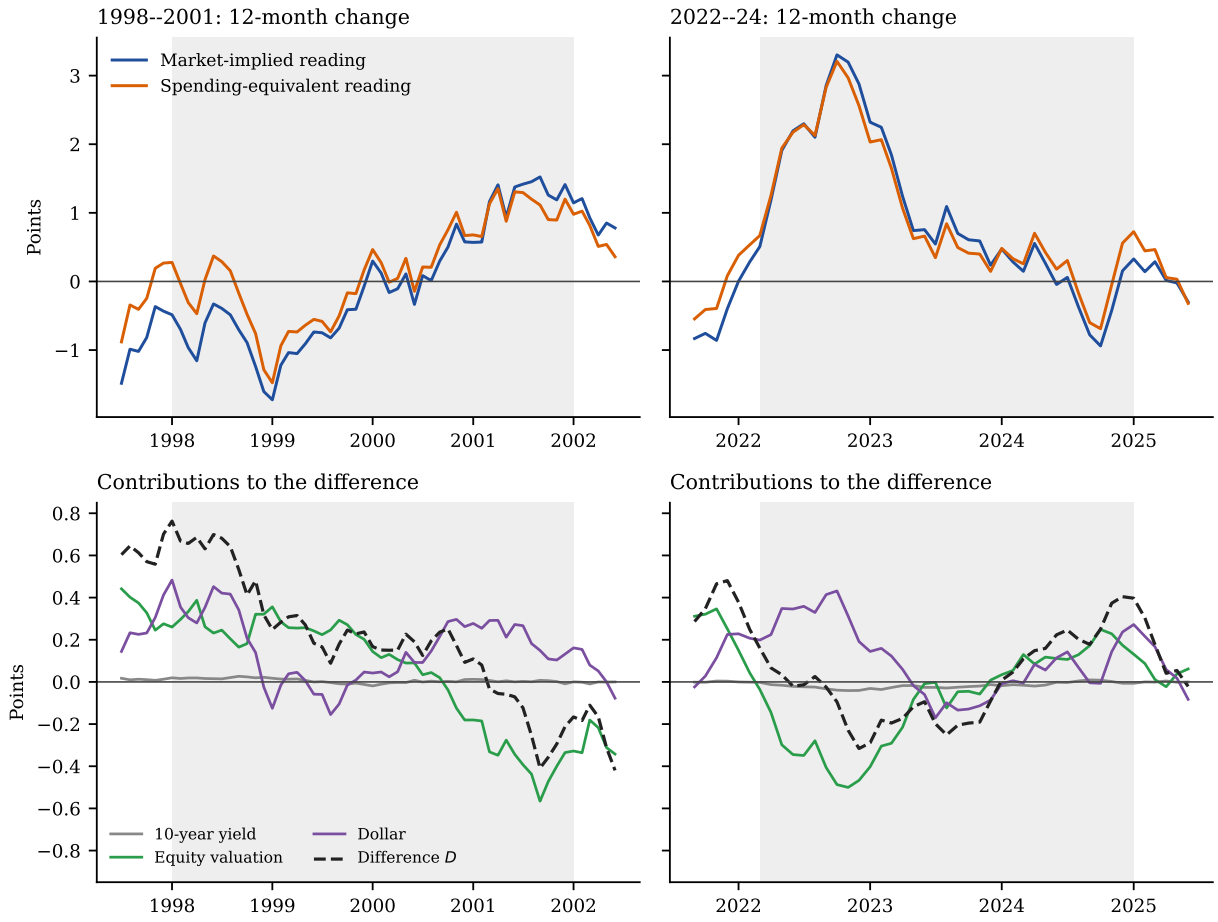


Figure 5: Month by Month: Readings and the Sources of Their Difference

Notes: Top: 12-month change of the market-implied and spending-equivalent readings on the same three assets relative to r^*. Bottom: each asset's contribution to D, the weight difference times its 12-month change, and D itself (dashed). The shaded area is the episode.

Table 6: How Often the Four Readings Give Different Judgments

	Percent of months	90% interval	Months
All four readings have the same sign	81.6	[70.2, 82.9]	301
At least two materially opposing readings, $t = 10$ bp	10.3	[8.4, 19.8]	38
At least two materially opposing readings, $t = 25$ bp	2.2	[1.6, 8.1]	8
Four-reading range at least 10 bp	92.4	[87.5, 97.6]	341
Four-reading range at least 25 bp	70.5	[58.5, 88.9]	260
Market (futures factor) and FRB/US materially opposing, $t = 25$ bp	2.2	[0.0, 5.4]	8

Notes: Monthly 12-month changes in the three assets relative to r^*, January 1996 to September 2026 (369 months, overlapping), read with the four portfolios on the common two-year-yield scale. Materially opposing: at least one reading at least t above zero and at least one at least t below zero. Four-reading range: the largest less the smallest of the four readings. Intervals: 5th and 95th percentiles over 5,000 bootstrap draws of the announcements (market weights and the common response vector redrawn; FRB/US and GS weights fixed), conditional on the observed history.

4.3 The Full History

A policymaker reading financial conditions wants to know two things: whether conditions tightened or eased, and by how much. Table 6 answers both across the four calibrations. All four readings of the 12-month change have the same sign in 81.6 percent of the 369 months (90 percent interval 70.2 to 82.9). At least two give materially opposing readings, one at least 25 basis points above zero and one at least 25 basis points below, in 8 months (2.2 percent; 1.6 to 8.1). With a threshold of 10 basis points the share rises to 10.3 percent. Both thresholds belong in view, because what counts as a meaningful disagreement is the policymaker's choice. The size of the change is less robust. The four-reading range, the largest reading less the smallest, is at least 25 basis points in 70.5 percent of months. That figure answers the question of a policymaker choosing among four calibrations; for one fixed pair, the futures-factor market index and FRB/US, the readings differ by at least 25 basis points in 36.6 percent of months (10 to 70; Table C.2 in the appendix). The intervals are conditional on the observed history and on the fixed FRB/US and GS weights. The rarity of materially opposing readings is the more informative result; how often magnitudes differ is estimated much less precisely.

The exceptions. Table 7 lists the 8 months of materially opposing readings. They fall in four periods, 1997–98, 2003, 2008 and 2015, and show two patterns. In 6 months, in late 1997, the first half of 1998 and early 2015, FRB/US alone reads a tightening of at least 25 basis points while the others read an easing. In each of those months the dollar had risen by more than 10 percent over the year, valuations had also risen by more than 10 percent and the 10-year yield had fallen. FRB/US gives the dollar the largest weight relative to equities, so the appreciation dominates its reading; the others give more weight to the rise in valuations and the fall in yields. In the other 2 months, May 2003 and June 2008, the futures-factor and GS readings are tight while FRB/US is easy. Valuations had fallen by more than 20 percent over the year and the dollar had depreciated, and those two portfolios weight equities most heavily relative to the dollar.

Table 7: The Months of Materially Opposing Readings

Month	Market, futures factor	Published GS weights	Market, two-year yield	FRB/US
Dec 1997	-0.43^{-}	-0.40^{-}	-0.24	0.27^{+}
Jan 1998	-0.49^{-}	-0.41^{-}	-0.42^{-}	0.28^{+}
Jun 1998	-0.33^{-}	-0.27^{-}	-0.23	0.37^{+}
Jul 1998	-0.39^{-}	-0.35^{-}	-0.26^{-}	0.29^{+}
May 2003	0.41^{+}	0.53^{+}	-0.28^{-}	-0.30^{-}
Jun 2008	0.46^{+}	0.56^{+}	-0.18	-0.26^{-}
Mar 2015	-0.37^{-}	-0.30^{-}	-0.37^{-}	0.26^{+}
Apr 2015	-0.37^{-}	-0.30^{-}	-0.35^{-}	0.25^{+}

Notes: Twelve-month change in financial conditions under each portfolio, in points on the common two-year-yield scale (normalized to the futures-factor response), in the months in which at least one reading is at least 0.25 above zero ($^{+}$) and at least one at least 0.25 below zero ($^{-}$).

5 What Can Subsequent Activity Establish?

5.1 The Test

Consider two years in which the market index tightened by the same amount. If the dollar accounted for more of the tightening in one year, did activity subsequently weaken more? If the market-implied reading is a sufficient summary of the asset prices, the answer is no. If the spending-equivalent reading is, the answer is yes, by the amount that the model's weights imply. I regress the 12-month-ahead change in the unemployment gap on own lags and

$$y_{t+12} - y_t = \cdots + \beta M_t + \gamma D_t, \tag{4}$$

where M is the 12-month change in the market index (relative to r^*) and D the 12-month change of the normalized FRB/US-minus-market difference on the same changes. If the market-implied reading is sufficient, $\gamma = 0$. If the spending-equivalent reading $M + D$ is sufficient, $\gamma = \beta$. I read the two restrictions as tests of incremental predictive sufficiency in the span of the two portfolios, conditional on the controls. Three assets span a larger space, and an unrestricted regression on the three assets is a different test. Failure to separate the portfolios for aggregate unemployment or GDP is consistent with low power and with forward-looking asset prices, and it cannot validate either structural weight. That wording matters for unemployment, since the FRB/US weights are constructed from effects on GDP; four-quarter GDP growth is the model-facing check. Table 8 reports both restrictions with the joint covariance of the estimates. The test compares two of the four calibrations, the futures-factor market reading and the FRB/US reading.

Table 8: Can Realized Activity Tell the Two Weightings Apart?

	n	Market index β [90%]	Composition γ [90%]	$\gamma - \beta$ [90%]
Unemployment gap, 12 months: 1976–2026	556	0.19 [0.05, 0.33]	−0.26 [−0.94, 0.43]	−0.45 [−1.01, 0.11]
Unemployment gap, 12 months: 1996–2026	318	0.01 [−0.21, 0.22]	−1.19 [−2.50, 0.13]	−1.19 [−2.31, −0.07]
Unemployment gap, 12 months: Excluding 2008–15	436	0.29 [0.15, 0.43]	0.32 [−0.13, 0.77]	0.03 [−0.30, 0.37]
Real GDP growth, four quarters: 1976–2026	186	−0.16 [−0.47, 0.16]	0.87 [−0.78, 2.51]	1.03 [−0.39, 2.44]
Real GDP growth, four quarters: 1996–2026	106	0.31 [−0.13, 0.75]	1.81 [−0.76, 4.38]	1.50 [−0.70, 3.71]
Real GDP growth, four quarters: Excluding 2008–15	146	−0.33 [−0.61, −0.04]	0.26 [−1.21, 1.72]	0.58 [−0.69, 1.86]

Notes: Local projections of the 12-month-ahead change in the unemployment gap (four-quarter-ahead GDP growth) on own lags and $y = \cdots + \beta M + \gamma D$, where M is the 12-month change of the market index relative to r^* and D the 12-month change of the normalized FRB/US reading less the market reading, computed on the same changes (HAC standard errors, 18 lags, joint covariance of the coefficients). If the market reading is right, $\gamma = 0$. If the normalized FRB/US reading $M + D$ is right, $\gamma = \beta$. A failure to reject is not evidence for either.

5.2 The Answer

On 1976–2026 the coefficient on the index change is 0.19 and on composition −0.26 (−0.94 to 0.43); the difference $\gamma - \beta$ is −0.45 (−1.01 to 0.11). Neither restriction is rejected. Moving D from its 5th to its 95th percentile at the same index change shifts the year-ahead outlook by −0.29 percentage point, with an interval from −1.06 to 0.48, so the sample cannot establish an outlook difference of either sign. On 1996–2026 the coefficient on the index change is 0.01 (−0.21 to 0.22) and on composition −1.19 (−2.50 to 0.13) and the model's restriction is rejected ($\gamma - \beta = -1.19$, −2.31 to −0.07); excluding 2008–15 the coefficient on composition is 0.32 and $\gamma - \beta$ is 0.03, close to the model's restriction. The sign changes across samples. For GDP growth neither restriction is rejected, and adding D does not significantly improve out-of-sample forecasts (Clark–West −1.8).

What can a policymaker take from this? The two readings can be used interchangeably when markets move as they do around policy news, and the data cannot say which to prefer when they do not. A reading from either index should be accompanied by the question of what moved: the dollar, valuations, or yields. The answer indicates whether the two readings are likely to part.

6 How Stable Are the Weights, and What Are the Limits?

6.1 Weights That Move

The weights define the unit, so I hold them fixed and show how they would drift. Re-estimating on an expanding window of announcements from 2006 keeps the 10-year and dollar weights inside the 90 percent band of the full-sample estimate at every date after 2010; the equity weight is outside the full-sample 90 percent bootstrap interval in 32 monthly estimates between January 2011 and January 2015, by at most −0.073 against a band edge of −0.068 (Figure 6). The overlapping monthly estimates are not 32 independent tests of instability. A 20-year rolling window fails after 2021, because it drops the announcements of 1996–2005. The 10-year weight is imprecise throughout. Apart from the equity weight in these years, the weights stay within their sampling error.

The other policy-news series and event samples are in Appendix D.1.

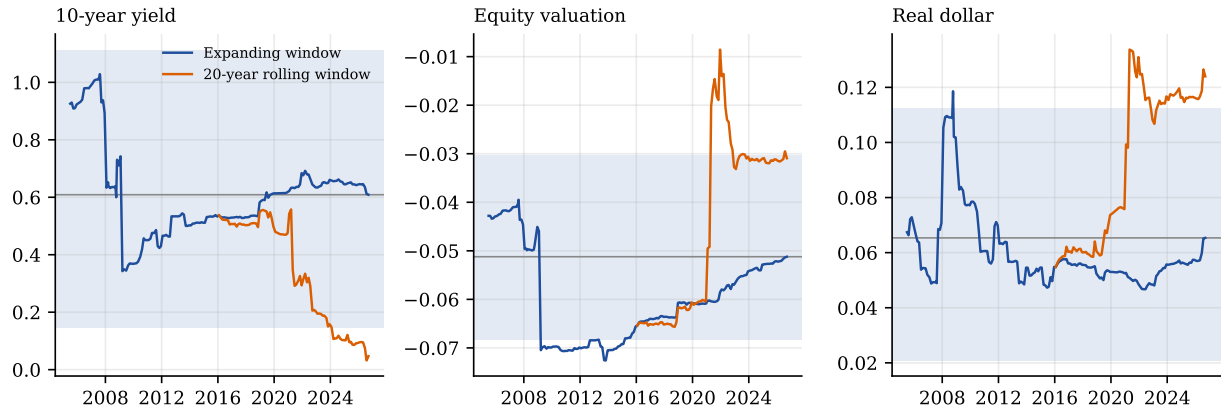


Figure 6: The Weights on an Expanding and a Rolling Window

Notes: Weights of the market index re-estimated on announcements to each date (expanding window, from the 80th announcement) and on the latest 20 years (rolling). Shaded: 90 percent bootstrap interval of the full-sample weights; the horizontal line is the full-sample weight.

6.2 Limits of the Design

The weights come from announcements since 1996 and are applied from 1973; they assume that the responses reflect policy rather than information effects (Nakamura and Steinsson, 2018), and if policy came to target financial conditions the responses could change, the Lucas-critique concern that Hatzius et al. (2010) raise for all indexes. The FRB/US weights come from one model, at one horizon and in one experiment, and are held fixed in the intervals. The daily comparison of Section 4.1 uses nominal assets, with the nominal broad dollar for the real dollar, no adjustment for r^* and weights fitted on the same announcements; the intraday comparison holds each event out. The sample has five recessions, two of them out of sample, and the 12-month outcomes overlap, so the counts of tests are descriptive and not adjusted for their number. The neutral rate is not neutral to outcomes, because the Laubach–Williams model uses output, inflation and unemployment.

7 Conclusion

The paper asks how many points of monetary policy a movement in financial conditions is worth. An event study says how large a conventional policy surprise the movement resembles, and a spending calibration says how large a policy-rate change would have a comparable effect on GDP. The two questions are different. Knowing which one an index answers, however, tells a reader surprisingly little about which other index it will resemble. On the movements of Figure 1 the futures-factor market index and the published Goldman Sachs weights give nearly the same relative readings, and the two-year-yield market index and FRB/US both give the appreciation more than twice the reading of the equity decline.

Across four calibrations put on one scale, financial conditions usually give the same directional

signal. Over 1996–2026 all four readings of the annual change have the same sign in 81.6 percent of months, and materially opposing readings occur in 8 of 369 months at a threshold of 25 basis points. The size of the tightening is more sensitive. Switching between two implementations of the same method can change the reading as much as switching from market responses to spending effects, and the two closest readings in the sample come from different methods, on the paper's series and on series closer to the GS inputs. In 2008–09 every calibration showed a large tightening and the size differed by more than half a point; in 2022–24 the weights differed and the readings nearly converged. The weights differ for reasons the paper can trace: the measure of policy news on the market side, and the trade channel of the dollar in FRB/US. Why the published Goldman Sachs weights match the futures-factor market weights is open.

GDP growth provides little evidence for choosing between the futures-based market reading and the FRB/US reading, and the unemployment comparison is sensitive to the sample. The FRB/US weights come from one model and one experiment and the published weights are applied to series they were not built on, so the four portfolios are a defined set of U.S. calibrations and not the population of indexes. For a policymaker, the practical reading is that the direction of a change in financial conditions is a robust signal across reasonable ways of translating it into policy-rate points, and its size is not. A reading of a quarter point of equivalent tightening should come with the calibration that produced it and with what moved: the dollar, valuations or yields.

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Appendix A The Weights: a Derivation

Let the movements in the three asset prices be $x = \psi s + u$, where s is policy news, ψ the responses and u everything else, with covariance Σ. Let g_h be the effects of a persistent unit shift in each asset price on spending at horizon h. The FRB/US weights are proportional to g_h. The market weights solve the problem of recovering s from x: the generalized least squares estimator is $\hat{s} = w_M' x$ with $w_M = \Sigma^{-1}\psi/(\psi'\Sigma^{-1}\psi)$, which satisfies $w_M'\psi = 1$ and, by the Gauss–Markov theorem, minimizes the variance of the estimate among linear unbiased estimators of s when u has mean zero and covariance Σ and is uncorrelated with s. For any x the difference between the normalized readings is $D_h(x) = (w_{B,h} - w_M)'x = (w_{B,h} - w_M)'u$, because the policy part cancels. The readings coincide for every movement if and only if $w_{B,h} = w_M$, that is, if $g_h \propto \Sigma^{-1}\psi$: the criterion that recovers policy news precisely and the criterion that measures spending effects select the same weights only in that case. The variance of the difference is $(w_{B,h} - w_M)'\Sigma(w_{B,h} - w_M)$.

Observed movements need not be of the form $\psi s + u$ with a single policy factor, and the model's effects need not be those of a movement of the same shape. If all observed movements lay along the single direction ψ, every weight vector normalized to price ψ at one would give the same reading on them. The data would then identify the reading along that direction and leave the weights themselves undetermined. In the announcement windows u is not zero, and it is the variation in u that identifies Σ and hence the weights. What distinguishes the two weightings in practice is the exposure of the difference in weights to the movements that occur, which Table A.1 measures as a ratio to the market move.

Appendix B Construction of the Series

The three asset prices enter in three forms, each answering a problem of the one before. The headline results use the third.

Nominal. Rates as published. A nominal index treats a 10-year yield of 8 percent alike in 1981 and in 2000. That is reasonable only if inflation expectations and the neutral real rate are stable,

Table A.1: How Large is the Difference Between the Two Readings?

Movements	Mean $\lvert D\rvert$	Ratio to market move [90%]	Share above 0.25 (%)	Mean run (months)
Announcement windows, held out	0.010	0.19 [0.09, 0.37]	0	–
Monthly changes, 1996–2026	0.043	0.24 [0.14, 0.54]	0	1.0
Monthly changes, 1973–2026	0.063	0.27 [0.15, 0.51]	2	1.1
12-month changes, 1996–2026	0.26	0.33 [0.17, 0.63]	37	7.5
12-month changes, 1973–2026	0.33	0.32 [0.16, 0.54]	45	7.3

Notes: D is the FRB/US reading (four quarters, normalized so that a conventional policy surprise reads one) less the market reading, applied to the changes of the equity valuation, the 10-year yield and the dollar at the r^* step. Announcement windows: held-out events (weights re-estimated without each event). Ratio: mean $\lvert D\rvert$ over mean absolute market move; interval from bootstrap samples of the announcements and, for the monthly history, of 12- or 24-month blocks of months. A run is consecutive months with $\lvert D\rvert$ above 0.25.

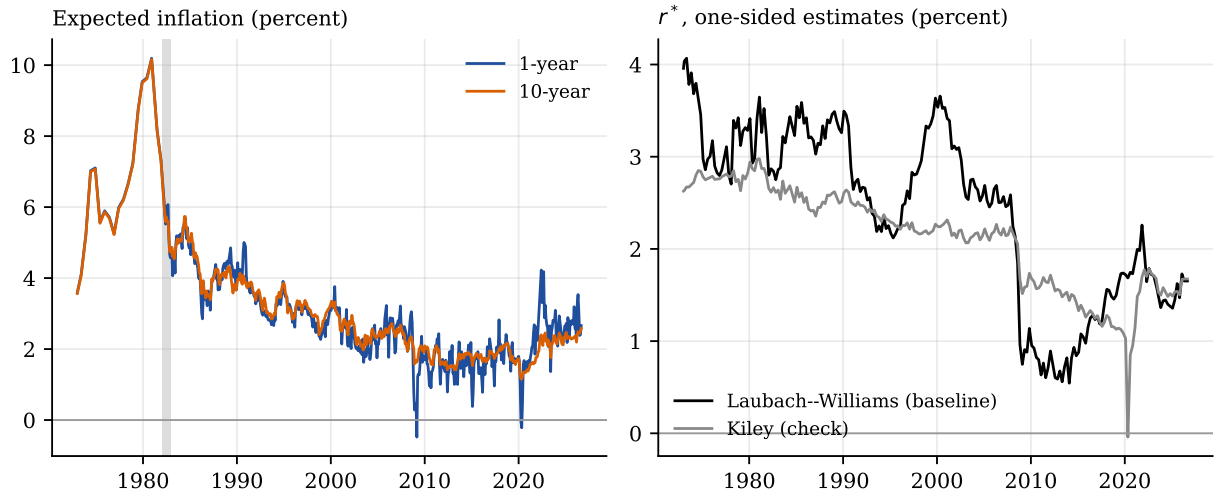


Figure B.1: Expected Inflation and r^*

Notes: Left: expected inflation, Livingston Survey before 1982 and Cleveland Fed model estimates after, shifted to the Cleveland level over 1982–84 and blended over 1982; shaded: the blending year. Right: one-sided, full-sample estimates of r^* from the Laubach–Williams model (baseline) and Kiley's model (check). Percent.

and neither was. Expected inflation over the next year averaged 8.43 percent in 1981 and is 2.66 percent today, and the Laubach–Williams estimate of r^* averaged 2.76 percent over 1990–2007 and 0.86 percent over 2009–2016 (Figure B.1).

Real. Each rate less expected inflation at a matched horizon: ten-year expected inflation for the 10-year yield. The dollar is the real broad index.

Relative to r^*. The real 10-year yield less r^*. Equity valuation enters through the earnings yield $E/P = 100/\text{CAPE}$, which is also an implicit real rate, as the gap $E/P - r^*$, converted to CAPE log points at the sample-average earnings yield so that its weight on CAPE is unchanged. The baseline uses the one-sided, full-sample Laubach–Williams estimate. Each variable is measured relative to its 1973–2026 average, so r^* matters through its movements over time.

The transformation matters for levels and not for the difference between the two readings. The ratio of the mean absolute difference to the mean absolute market move in 12-month changes, 1996–2026, is 0.33 (interval 0.18 to 0.68) with nominal assets and 0.33 relative to r^*.

Data. All series are monthly averages from January 1973 to September 2026, from the public sources in Table B.1. The BBB spread is Moody's Baa yield less the 10-year yield. Months in which the CPI and unemployment releases are missing (October 2025) are interpolated. The unemployment gap is the unemployment rate less the Congressional Budget Office's natural rate, placed in the middle month of each quarter and interpolated. Expected inflation is the Cleveland Fed's estimate at one and ten years from 1982 and the Livingston Survey's 12-month CPI expectation before, shifted to the Cleveland level by the mean difference over 1982–84 and blended over 1982. CAPE is Shiller's series spliced over 1990–92 in logs to the monthly average of a daily series. The real broad

dollar is spliced in 2006 in logs, as are the nominal dollar and the S&P 500.

Market index. Policy news is the first principal component of MP1, MP2, ED2, ED3 and ED4 in the monetary-event windows, scaled by its regression on the two-year yield change so that one unit moves the two-year yield by one percentage point. Unscheduled events with no change in MP1 and March–April 2020 are dropped. Each of the 10-year yield change, the S&P 500 return and the dollar (DXY) return in the window is regressed on the factor. The residual covariance Σ of the three regressions gives $w = \Sigma^{-1}\psi/(\psi'\Sigma^{-1}\psi)$. The S&P 500 and DXY weights apply to 100 times the log of CAPE and of the real broad dollar, as percent changes in the window and in the index levels agree to first order. The weights bootstrap resamples announcements with replacement and recomputes the policy-news factor and the weights each time; draws use the seed in `lib/config.py`.

FRB/US index. The weights are the cumulative GDP responses at four quarters to a persistent unit shift in each variable, simulated in FRB/US with VAR-based expectations, divided by the response to a persistent one percentage point rise in the funds rate. The weight on the policy rate is its direct effect with expectations of future rates held fixed. House prices enter as 100 times the log of the price-to-rent ratio relative to its trailing 15-year average. The expenditure decomposition of Figure 3 uses the model's own identities for consumption, residential and business fixed investment, inventories, government, exports and imports, weighted by their shares in GDP, with a residual below 10^{-7} percentage points; it is run with the Board's FRB/US package and the data vintage of October 2026, downloaded by step 00 into a gitignored directory, with a Python 3.11 environment built on first use. The policy-rate experiment shocks the inertial-Taylor-rule equation in the first quarter and is scaled (to 2.14 units of the add factor) so that the average simulated funds rate over eight quarters rises by one point; GDP at four quarters is linear in the size of the shock to within 1.6 percent. For the decay experiments of Section C.4, each variable's shock decays geometrically with the half-life stated, and the response is computed in the model by linearity from the temporary-shock responses.

Policy-news definitions. The two-year definition replaces the futures factor by the change in the two-year Treasury yield in the same window, with the same regressions and the same bootstrap of announcements; the draws for different definitions are paired by keeping, for each resample, the announcements that fall in each subsample, so the subsample size varies by draw. The announcement bootstrap uses 5,000 draws for every weights-based interval; the block bootstraps of the VAR and the outcome regressions use 500. The FRB/US weights are held fixed.

Forecasting data. The regressions use local projections of the 12-month-ahead change in the unemployment gap (four-quarter-ahead GDP growth) with HAC standard errors of 18 lags.

Appendix C The Market and FRB/US Pair

C.1 Across Horizons

The pair that the comparison usually features, the futures-factor market index and FRB/US, can be followed at horizons from an announcement window to a year. On the 253 announcements, with

Table B.1: Data Sources

Series	Frequency	Source
Federal funds rate; 10-year yield; Baa yield; 30-year mortgage rate	Monthly	FRED FEDFUNDS, GS10, BAA, MORTGAGE30US
Real broad dollar; nominal broad dollar	Monthly	FRED RTWEXBGS, TWEXBPA; DTWEXBGS, TWEXBMTH (spliced in 2006)
CPI; CPI rent; Case–Shiller home prices; S&P 500	Monthly	FRED CPIAUCSL, CUUR0000SEHA, CSUSHPISA, SP500; Shiller's data before the FRED start
Unemployment rate; natural rate of unemployment; real GDP	Monthly; quarterly	FRED UNRATE; CBO, FRED NROU; GDPC1
Expected inflation, 1 and 10 years	Monthly	Cleveland Fed, FRED EXPINF1YR, EXPINF10YR; before 1982, Livingston Survey (Philadelphia Fed)
r^*, Laubach–Williams	Quarterly	New York Fed, one-sided estimates
r^*, Kiley	Quarterly	One-sided estimates of the author
CAPE	Monthly	Shiller, spliced to the monthly average of a daily CAPE constructed by the repository's daily workflow
FRB/US simulations	Quarterly	Run for this paper's companion work with the public FRB/US package
Futures, yields, S&P 500 and DXY in the window	Event	U.S. Monetary Policy Event-Study Database, "monetary events" (Acosta et al., 2025)

the weights re-estimated without the event being evaluated, the ratio in intraday windows is 0.19 (interval 0.09 to 0.37), and the 90th percentile of the difference is 0.023 against a mean absolute market move of 0.053. These are the windows in which the weights are estimated, with each event held out.

The ratio is larger in daily data than in the intraday windows and does not shrink at longer horizons. On daily changes of the same three assets, 1996–2026, it is 0.30 on announcement days and 0.32 on other days; at one week it is 0.27 and 0.29, and at 22 trading days 0.27 and 0.25 (Table C.1). No difference between announcement and other windows is distinguishable from zero. Announcement days contain a great deal of financial news besides the announcement. The fitted response to policy news is a particular combination of asset prices, and the difference over a whole announcement day looks like the difference on other days.

For windows that end on any day, the ratio is 0.32, 0.29, 0.27, 0.30 and 0.34 at 1, 5, 22, 66 and 252 trading days. A window of 22 or more days contains announcements in any case; removing the moves on announcement days gives 0.33, 0.29, 0.28, 0.32 and 0.35, within 0.02 of those. In 12-month changes of the monthly series, the ratio is 0.33 (interval 0.17 to 0.63), and the difference exceeds 0.25 in 37 percent of months (Figure C.1). The months overlap, so runs in this series are not independent evidence of persistence. The statistic is a sample summary with wide intervals. Roughly 0.25 to 0.33 describes the baseline across horizons from a day to a year, and the ratio is 0.36 to 0.42 under the other definitions of policy news (Table D.3).

Table C.1: The Difference Between the Readings on Announcement Days and Other Days

Horizon	Announcement windows [90%]	n	Other windows [90%]	n
One day	0.30 [0.17, 0.56]	251	0.32 [0.18, 0.61]	7339
One week	0.27 [0.16, 0.57]	251	0.29 [0.16, 0.59]	1265
22 trading days	0.27 [0.15, 0.56]	248	0.25 [0.14, 0.52]	98

Notes: Mean absolute difference D between the normalized FRB/US reading and the market reading, as a ratio to the mean absolute market move, on the same three assets, 1996–2026 (daily 10-year yield, nominal broad dollar and CAPE; the nominal dollar and no adjustment for r^* stand in for the real dollar and the r^* transformation). Announcement windows end on a scheduled FOMC announcement date; other windows are non-overlapping and contain none. Intervals from joint bootstrap samples of the windows and of the announcements that set the weights.

Sampling noise. How large is that relative to measurement noise in the market weights themselves? The same statistic computed on bootstrap draws of the baseline market weights against their point estimates has a median of 0.25 (5th to 95th percentile 0.08 to 0.51), and the baseline market weights against those of the two-year definition give 0.37 (0.15 through 2019 and 0.28 without the lower-bound years), against 0.33 for the spending-equivalent weights against the baseline market weights. The disagreement between the model and the market is therefore about as large as the disagreement between two market calibrations that differ only in the measure of policy news (0.37 against 0.33), and it is of the size of the sampling noise in the market weights. The sampling noise limits how finely the two calibrations can be told apart. It is a statement about precision and is separate from the differences across calibrations.

The intraday and daily figures do not isolate a source for the gap between 0.19 and 0.3. The intraday windows are narrow and the weights exclude the event evaluated. The daily comparison uses nominal assets and weights fitted on the same announcements (Appendix D). The comparison across horizons is the cleaner result: the disagreement does not shrink as the window lengthens.

C.2 Months, Episodes and Levels

Table C.2 counts the months in which the two readings of the pair agree or differ, Table C.3 gives their readings of three episodes in 12-month changes and in levels, and Table C.4 accounts for the difference in average levels between the full FRB/US index and the market index, which also differ in coverage (the policy rate, spreads and house prices) and centering.

C.3 Trade-Sensitive and Domestic Expenditure

Section 3.3 traced the model's weights to channels: consumption and investment for equities, net exports for the dollar. If the weights describe those channels, composition might predict the components separately even where it does not predict the aggregates. I wrote two predictions from the channel logic before estimating the components: the coefficient on D is negative for the contribution of net exports to growth over the next four quarters (P1), and it is larger for consumption and private investment than for net exports (P2). The aggregate nulls of the preceding subsections

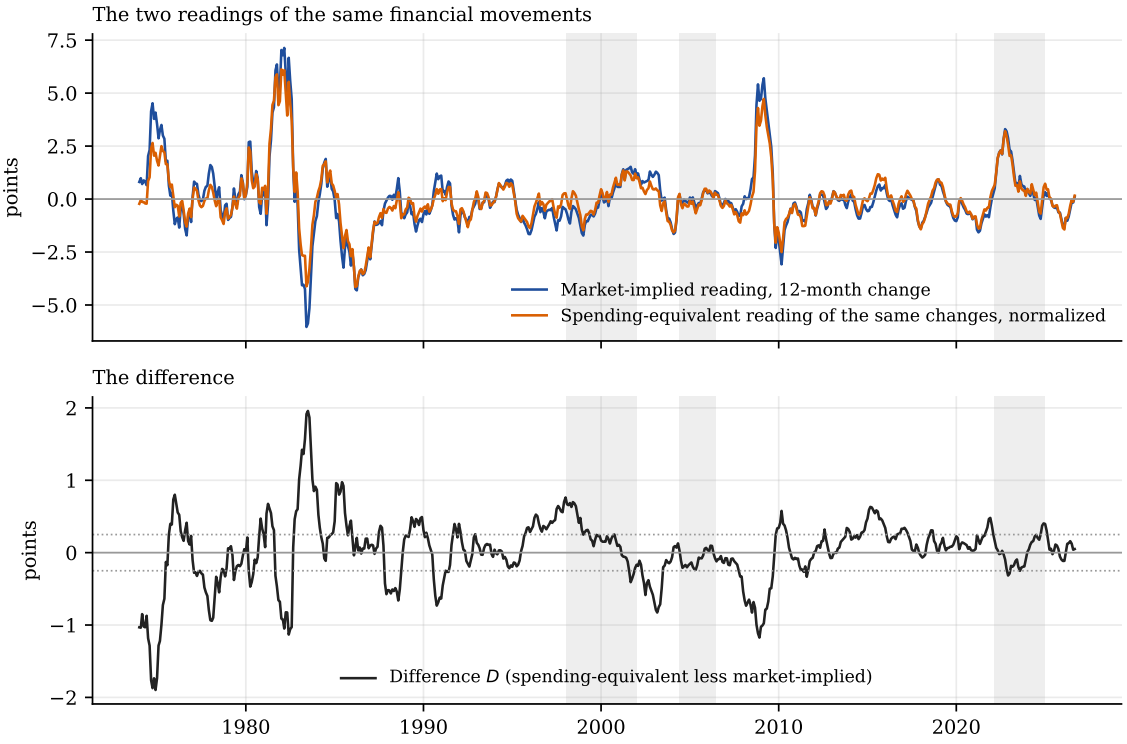


Figure C.1: Two Readings of the Same Financial Movements

Notes: Monthly, 12-month changes of the equity valuation, the 10-year yield and the dollar relative to r^*: the 10-year yield less ten-year expected inflation less r^*, the earnings-yield gap to r^* converted to CAPE log points, and the real dollar (Appendix B). Top: the change in the market index and the spending-equivalent reading of the same changes, normalized so that a conventional policy surprise reads one. Bottom: the difference D, FRB/US less market, with dotted lines at ±0.25. Shaded: 1998–2001, 2004–06 and 2022–24.

Table C.2: How Often the Two Readings Differ

	1996–2026	1998–2001	2022–24
Months	369	48	34
Threshold t = 0.10 point			
Same sign	323 (87.5%) [78, 94]	41 (85.4%) [75, 94]	33 (97.1%) [76, 100]
Opposite signs, at least one reading within t of zero	27 (7.3%) [5, 10]	3 (6.2%) [4, 12]	1 (2.9%) [0, 12]
Opposite signs, both readings at least t	19 (5.1%) [0, 13]	4 (8.3%) [0, 17]	0 (0.0%) [0, 15]
Readings differ by at least t	269 (72.9%) [48, 88]	41 (85.4%) [54, 96]	23 (67.6%) [35, 91]
Threshold t = 0.25 point			
Same sign	323 (87.5%) [78, 94]	41 (85.4%) [75, 94]	33 (97.1%) [76, 100]
Opposite signs, at least one reading within t of zero	38 (10.3%) [6, 17]	4 (8.3%) [6, 19]	1 (2.9%) [0, 21]
Opposite signs, both readings at least t	8 (2.2%) [0, 5]	3 (6.2%) [0, 6]	0 (0.0%) [0, 3]
Readings differ by at least t	135 (36.6%) [10, 70]	21 (43.8%) [15, 81]	6 (17.6%) [0, 74]

Notes: Monthly 12-month changes in the three market variables (relative to r^*), read with the baseline market weights and with the FRB/US weights on the market's scale (points of the two-year yield). Each cell gives the number of months, the share of months, and the 5th to 95th percentile of the share over announcement-bootstrap draws of the market weights (the 12-month changes are held fixed; months overlap). 1996–2026 is 369 months, 1998–2001 is 48 and 2022–24 is 34.

Table C.3: Two Readings of Three Episodes

	1998–2001	2004–06	2022–24
Full indexes, average level (points)			
FRB/US index less market index [90% interval]	1.79 [0.33, 2.97]	−0.11 [−0.76, 0.41]	0.96 [0.00, 1.76]
of which: equity weights	1.32	0.67	0.54
Same three assets, 12-month change (points)			
Market-implied reading	−0.06	0.04	1.02
Spending-equivalent reading	0.16	−0.05	1.01
Difference D	0.22	−0.09	−0.01
Change in the unemployment gap over the next 12 months (points)			
Realized	0.35	−0.41	0.27
Forecast from the market-implied reading	0.67	0.13	0.65
Forecast from the spending-equivalent reading	0.78	0.10	0.66

Notes: Top block: difference in the average levels of the full FRB/US index (with house prices) and the market index relative to r^*, in points, and the part that comes from the equity weights (Table C.4). Middle block: 12-month changes of the market index and of the normalized FRB/US reading, computed on the same three assets and averaged over the episode; D is their difference. Bottom block: change in the unemployment gap over the following 12 months, averaged over the episode's origins, and recursive out-of-sample forecasts of it from the market reading and from the FRB/US reading (own lags included). A gap in levels can coexist with agreement in changes.

Table C.4: From the Market Index to the FRB/US Index: Shares of the Difference

Stage	Level, share of variance (%)	12-month change (%)
Equity weights	72 [0, 89]	53 [-4, 87]
Dollar weights	12 [-1, 38]	15 [-1, 34]
10-year weight	1 [-4, 51]	1 [-7, 55]
Normalization of the weights	5 [-4, 19]	4 [-2, 16]
Policy rate	-0 [-1, 0]	0 [-1, 2]
BBB spread	1 [-1, 8]	3 [-3, 15]
Mortgage spread	2 [-3, 15]	15 [-0, 43]
House prices	7 [-2, 34]	8 [0, 24]

Notes: Shares of the variance of the FRB/US index (with house prices) less the market index, relative to r^*, 1973–2026, attributed to each stage of a bridge between them: the weights on the three common assets, the normalization of the weights, then the contributions that only the FRB/US index has (policy rate, spreads, house prices); both indexes use the same inputs and centering. Intervals (5th to 95th percentile) from bootstrap samples of the announcements and of blocks of months.

Table C.5: Composition and the Expenditure Components of Growth

	n	Coefficient [90% HAC]	Bootstrap 90%	Effect, 5th to 95th percentile
Net exports, 1976–2026: index change β	199	−0.11 [−0.19, −0.02]	[−0.19, 0.00]	−0.11
Net exports, 1976–2026: composition γ	199	−0.74 [−1.06, −0.41]	[−1.04, −0.37]	−0.79
Consumption and private investment, 1976–2026: index change β	199	−0.19 [−0.54, 0.15]	[−0.52, 0.22]	−0.21
Consumption and private investment, 1976–2026: composition γ	199	1.71 [−0.29, 3.70]	[−0.35, 3.92]	1.83
Consumption and investment less net exports, 1976–2026: γ	199	2.44	[0.13, 4.88]	2.63
Net exports, 1996–2026: index change β	119	0.03 [−0.13, 0.19]	[−0.13, 0.24]	0.03
Net exports, 1996–2026: composition γ	119	−0.54 [−1.17, 0.09]	[−1.28, 0.10]	−0.48
Consumption and private investment, 1996–2026: index change β	119	−0.31 [−1.10, 0.48]	[−1.13, 0.72]	−0.27
Consumption and private investment, 1996–2026: composition γ	119	2.95 [−0.65, 6.55]	[−0.98, 6.11]	2.59
Consumption and investment less net exports, 1996–2026: γ	119	3.49	[−0.99, 7.17]	3.07

Notes: Local projections of the mean contribution to real GDP growth (percentage points, annualized) of net exports and of consumption plus private investment over the next four quarters on own lags, the 12-month change of the market index and the 12-month change of D (Section 5.1); quarterly origins, HAC standard errors with 6 lags, 8-quarter block bootstrap with common draws for the two outcomes. Predictions stated from the channel logic before the components were estimated: the coefficient on D is negative for net exports (P1) and larger for domestic demand than for net exports (P2). Effect: coefficient times the 5th-to-95th percentile range of D at a given index change.

were already known, so the exercise is a check on a prediction and not a clean pre-registered test. Table C.5 reports both.

In 1976–2026 the coefficient on D for net exports is −0.74 (−1.06 to −0.41; bootstrap interval −1.04 to −0.37), an effect of −0.79 percentage points of growth between the 5th and 95th percentiles of D. For consumption and private investment it is 1.71 (−0.29 to 3.70), and the difference has a bootstrap interval of 0.13 to 4.88. Both predictions hold in the baseline main sample. They do not survive further scrutiny as evidence for the dollar channel. In this sample D is dominated by equities: it correlates 0.86 with the 12-month change in equity valuation and 0.31 with the change in the dollar. Splitting D into the parts due to each asset, the equity part has a coefficient of −0.97 (−1.67 to −0.27) and the dollar part −0.52 (−1.20 to 0.16). Rising equities strengthen domestic demand and imports, which lowers the net-export contribution mechanically. With the three asset changes entered freely, the dollar's own coefficient is negative, as the dollar channel implies (−0.030, −0.050 to −0.011), and so is the equity coefficient (−0.010). On 1996–2026 the signs are the same and the evidence is weaker: the net-export coefficient is −0.54 (−1.17 to 0.09). The result also depends on the definition of policy news (Table C.6): the coefficient on D for net exports is −0.74 for the baseline, −0.54 through 2019, −0.24 without the lower-bound years and 0.02 with the two-year yield, where it is indistinguishable from zero. Composition predicts net exports under the futures-factor definitions, mostly through equities. It does not provide clean evidence for the dollar channel of the model.

Table C.6 also repeats the unemployment and GDP regressions under each definition. The coefficient on composition is not significantly different from zero in any of them, and the model's restriction $\gamma = \beta$ is rejected at the 10 percent level for unemployment only without the lower-bound years.

Table C.6: Composition and Outcomes Under Each Definition of Policy News

	Unemployment, γ	$\gamma - \beta$	GDP growth, γ	$\gamma - \beta$	Net exports, γ
Futures factor	−0.26 [−0.94, 0.43]	−0.45 [−1.01, 0.11]	0.87 [−0.78, 2.51]	1.03 [−0.39, 2.44]	−0.74 [−1.06, −0.41]
Two-year yield	−0.10 [−0.50, 0.31]	−0.33 [−0.75, 0.09]	0.53 [−0.51, 1.57]	0.77 [−0.25, 1.78]	0.02 [−0.37, 0.41]
Factor, through 2019	−0.12 [−0.64, 0.39]	−0.31 [−0.71, 0.09]	0.54 [−0.67, 1.76]	0.71 [−0.29, 1.70]	−0.54 [−0.78, −0.30]
Factor, no lower bound	−0.19 [−0.57, 0.20]	−0.38 [−0.72, −0.03]	0.68 [−0.35, 1.72]	0.85 [−0.07, 1.77]	−0.24 [−0.55, 0.07]

Notes: Coefficient on the composition term D (and its difference from the coefficient on the market index change) in the regressions of Sections 5.1 and C.3, 1976–2026, with D and the market index change computed with the market weights of each definition of policy news and the model weights normalized by that definition's $b_4'\psi$; HAC 90 percent intervals.

C.4 Composition or Persistence?

The comparison confounds two things. A persistent dollar appreciation in FRB/US and the dollar's response to policy news are different experiments, so a difference in weights could reflect the initial composition of the movement or the path that follows it. Figure C.2 separates them with the same set of calculations under common decaying paths for the three variables, with half-lives of 8, 4 and 2 quarters.

Persistence changes how much tightening a policy move represents. The model's reading of the usual response falls from 0.954 with persistent paths to 0.82 (interval 0.73 to 0.92) with a half-life of four quarters and 0.71 with a half-life of two. Persistence does not change which movement counts more. With the same decay for the dollar and for equities, the model's dollar-to-equity ratio is 2.53, 2.55 and 2.59 at half-lives of 8, 4 and 2 quarters, against 1.00 for the market weights (interval 0.33 to 2.25). Different half-lives for the two assets give a wider range, 1.91 to 3.41, and none near one. Across horizons of two to eight quarters and these paths the model's reading of the usual response runs from 0.37 to 0.95. The persistent case is the closest to the data, since the half-lives of the levels of the three variables since 1996 are 20 to 26 quarters, so the range for decaying paths shows the sensitivity of the conversion and not a likely value. The contrast between the two weightings is a contrast in the effect on GDP per unit of each asset price. The definition of policy news (Section 3.1) moves the contrast by an amount at least as large.

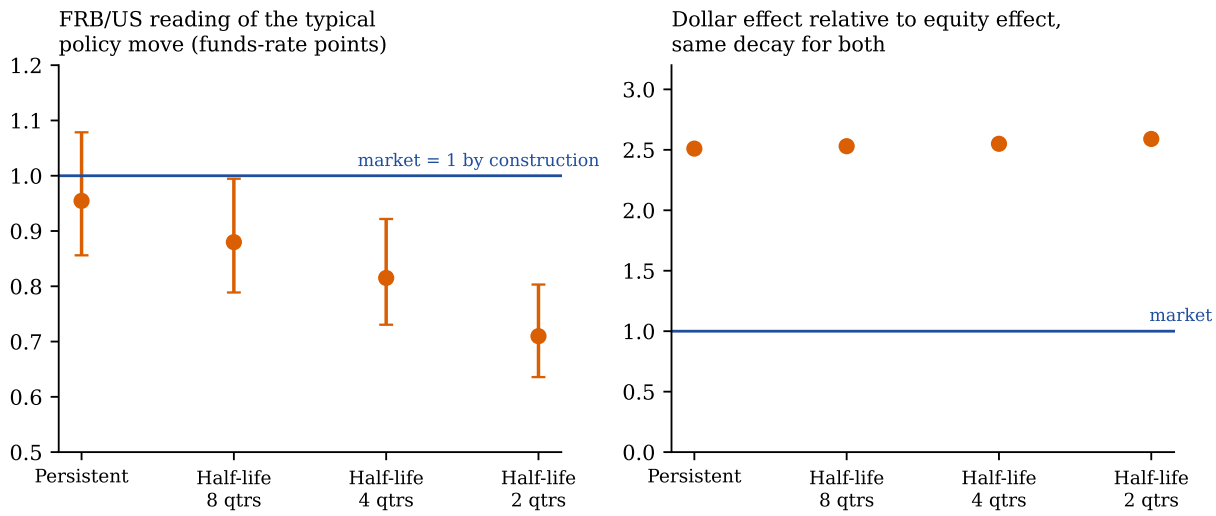


Figure C.2: Persistence Changes the Conversion, Not the Relative Weights

Notes: Left: the FRB/US weights at four quarters applied to the responses of the three variables to a unit of policy news, with the variables assumed to persist or to decay with the half-life shown; the bars are 90 percent intervals from bootstrap samples of the announcements. Right: the model's dollar effect over its equity effect, with the same decay for both; the line is the market index's ratio. The persistence of the variables' levels is a different object from the persistence of their responses to policy news; the half-lives of the levels since 1996 are 20 to 26 quarters, and the equity response to policy news does not decay within 20 trading days.

C.5 Tables of the Weights

Table C.7: Two Measures of Policy News on the Same Announcements

	Futures factor	Two-year yield
Announcements	253	253
Market weight, 10-year yield	0.609	0.925
Market weight, equity valuation	−0.051	−0.027
Market weight, dollar	0.065	0.070
Market dollar-over-equity ratio [90%]	1.00 [0.33, 2.25]	2.06 [0.82, 5.19]
Readings of the two movements of Figure 1		
Equity valuation falls 5 log points, market (points)	0.256	0.133
Dollar appreciates 3.9 log points, market (points)	0.255	0.274
Equity valuation falls, FRB/US (points, normalized for each signal)	0.160	0.164
Dollar appreciates, FRB/US (points, normalized for each signal)	0.402	0.413
Dollar-minus-equity effect, market (points)	0.00	0.14
Dollar-minus-equity effect, FRB/US (points, normalized for each signal)	0.24	0.25
Model less market [90%]	0.243 [0.003, 0.448]	0.107 [−0.079, 0.287]
Ratio of ratios [90%]	2.53 [1.08, 7.41]	1.22 [0.48, 3.05]
Model reading of the usual response, $b_4'\psi$ [90%]	0.954 [0.856, 1.078]	0.930 [0.834, 1.048]

Notes: The market weights are estimated on the same 253 announcement windows with two measures of policy news: the first principal component of the futures-based factor of Nakamura and Steinsson (2018) (baseline) and the change in the two-year yield. The model weights are those of Table C.9 in both columns, normalized by $b_4'\psi$ for each measure. Effects are for the movements of Figure 1, in points of the two-year yield; the FRB/US effects are divided by $b_4'\psi$ computed with each signal's response vector, so this table's normalization changes with the signal, unlike the common futures-factor normalization of Tables 1 and 3 (Table C.8 gives the FRB/US effects in funds-rate points). Intervals are 5th and 95th percentiles over 5,000 bootstrap samples of the announcements, on common draws. Table D.2 adds the two sample checks.

Table C.8: Two Financial Movements that the Market Index Rates Alike

	Market index	90% interval	FRB/US index
Equity valuation falls 5 log points	0.256	[0.152, 0.342]	0.153
Dollar appreciates 3.9 log points	0.255	[0.081, 0.438]	0.384
Dollar effect over equity effect	1.00	[0.33, 2.25]	2.51
Typical joint response to a unit of policy news	1.00		0.954 [0.86, 1.08]

Notes: Weights applied to the stated movements, with no change in the 10-year yield. Market index: two-year-yield points per movement, weights from announcement-window responses (Table 2); the interval takes the 5th and 95th percentiles over bootstrap samples of the announcements. FRB/US index: funds-rate points; the effect on GDP after four quarters of a persistent shift in the variable, divided by the effect of a persistent one percentage point rise in the funds rate. Typical joint response: the index applied to the responses of the three variables to a unit of policy news (one by construction for the market index).

Table C.9: FRB/US and Market Weights on the Three Common Variables

	FRB/US weight	Market weight	90% interval	FRB/US response	Market response
10-year yield (pp)	0.563	0.609	[0.147, 1.114]	0.37	0.53
Equity valuation (percent)	−0.031	−0.051	[−0.068, −0.030]	−13.87	−7.78
Real dollar (percent)	0.098	0.065	[0.021, 0.112]	1.56	4.22

Notes: FRB/US weight: effect on GDP after four quarters of a persistent unit shift in the variable, divided by the effect of a persistent 1 percentage point rise in the federal funds rate (0.4616 percent of GDP); funds-rate points per unit. Market weight: two-year-yield points per unit (Table 2); the interval is its bootstrap interval. FRB/US response: change in the variable four quarters after a persistent 1 percentage point funds-rate rise (VAR expectations). Market response: change in the announcement window per unit of policy news.

Table C.10: Agreement on Announcements and the Contrast in Composition

FRB/US horizon	Joint response to policy news [90%]	Normalized equity weight	Normalized dollar weight	Dollar over equity, FRB/US	Ratio of ratios [90%]
2 quarters	0.901 [0.81, 1.01]	−0.032	0.089	2.18	2.19 [0.94, 6.44]
4 quarters	0.954 [0.86, 1.08]	−0.032	0.103	2.51	2.53 [1.08, 7.41]
8 quarters	0.917 [0.82, 1.04]	−0.032	0.106	2.57	2.59 [1.11, 7.59]

Notes: FRB/US weights for GDP at the horizon shown, persistent-shift experiment, in funds-rate points; the joint response is the FRB/US weights applied to the responses to a unit of policy news (market index: one). Normalized weights: FRB/US weights divided by that response, so that both indexes read one on a conventional surprise; market weights are 0.609 (10-year yield), −0.051 (equity valuation) and 0.065 (dollar). Ratio of ratios: the FRB/US dollar-over-equity ratio divided by the market's (1.00); the interval comes from the bootstrap of the market weights, and the FRB/US weights are held fixed.

Appendix D Sensitivity

D.1 Other Policy-News Series and Event Samples

Table D.2 gives the weights, the readings of the two movements of Figure 1 and the contrasts under the two measures of policy news and the two sample checks, and Table D.1 re-estimates the weights with the two-year yield as the policy news, with announcements through 2019, and without the lower-bound years. The equity weight is −0.027, −0.061 and −0.039 against −0.051 in the baseline, and the conclusions about composition rest on the model's ratio of dollar to equity effects, not on any one set of market weights.

The episode readings of the futures-factor market index and FRB/US under each definition of policy news are in Table D.3.

Table D.1: Weights Under Other Policy-News Series and Event Samples

	Events	w 10-year	w equities	w dollar	ψ 10-year	ψ equities	ψ dollar
Baseline: Nakamura–Steinsson factor	253	0.609	−0.051	0.065	0.53	−7.78	4.22
Policy news is the 2-year yield change	253	0.925	−0.027	0.070	0.58	−6.09	4.22
Announcements through 2019	201	0.613	−0.061	0.053	0.56	−7.50	3.83
No announcements December 2008–November 2015	197	0.986	−0.039	0.058	0.49	−7.82	3.69

Notes: Weights w and responses ψ as in Table 2, re-estimated on each sample or with the window change of the two-year yield as the policy-news measure. In each case the weights applied to the responses sum to 1.

Table D.2: Policy-News Definitions and Event Samples

	Futures factor	Two-year yield	Factor, through 2019	Factor, no lower bound
Announcements	253	253	201	197
Market weight, 10-year yield	0.609	0.925	0.613	0.986
Market weight, equity valuation	−0.051	−0.027	−0.061	−0.039
Market weight, dollar	0.065	0.070	0.053	0.058
Model reading of the usual response, $b_4'\psi$ [90%]	0.954 [0.856, 1.078]	0.930 [0.834, 1.048]	0.919 [0.805, 1.091]	0.876 [0.790, 0.983]
Market dollar-over-equity ratio [90%]	1.00 [0.33, 2.25]	2.06 [0.82, 5.19]	0.68 [0.09, 2.11]	1.16 [0.38, 2.89]
Ratio of ratios [90%]	2.53 [1.08, 7.41]	1.22 [0.48, 3.05]	3.70 [0.84, 16.30]	2.16 [0.85, 6.16]
Dollar-minus-equity effect, market (points)	0.00	0.14	−0.10	0.03
Dollar-minus-equity effect, FRB/US (points, normalized for each signal)	0.24	0.25	0.25	0.26
Model less market [90%]	0.243 [0.003, 0.448]	0.107 [−0.079, 0.287]	0.349 [0.036, 0.565]	0.232 [0.021, 0.405]
Mean $\lvert D\rvert$ over mean |market move|, 12-month changes	0.33	0.36	0.42	0.39

Notes: The market weights are re-estimated for each definition of policy news: the first principal component of the futures-based factor of Nakamura and Steinsson (2018) (baseline), the change in the two-year yield in the same 253 announcement windows, the factor through 2019, and the factor outside the lower-bound years. The model weights are those of Table C.9 in every column, normalized by $b_4'\psi$ for each definition. Effects are for the movements of Figure 1 (equity valuation down 5 log points, dollar up 3.9 log points), in points of the two-year yield; the FRB/US effects are divided by $b_4'\psi$ and so are on the common two-year-yield scale. Intervals are 5th and 95th percentiles over 5,000 bootstrap samples of the announcements, on common draws.

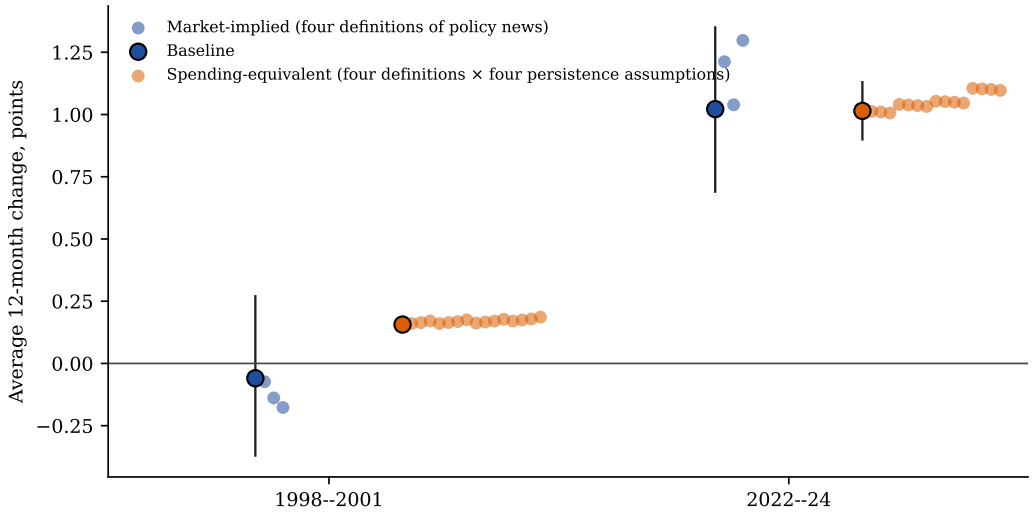


Figure D.1: Two Readings of Two Episodes under Every Measurement Choice

Notes: Average 12-month change over the months of the episode on the same three assets relative to r^*. Market-implied: the four definitions of policy news of Table D.2, in points of the two-year yield. Spending-equivalent: the same four definitions, which enter through the normalization, crossed with persistent paths and common half-lives of 8, 4 and 2 quarters; the FRB/US readings are divided by the model's reading of each definition's policy response and so are in points of the two-year yield, normalized for each definition. Large markers are the baseline. These are retrospective estimates.

Table D.3: Episode Readings Under Each Definition of Policy News

	Futures factor	Two-year yield	Factor, through 2019	Factor, no lower bound
1998–2001, average 12-month change				
Market-implied reading	−0.06	−0.07	−0.14	−0.18
Spending-equivalent reading	0.16	0.16	0.16	0.17
Difference D	0.22	0.23	0.30	0.35
2022–24, average 12-month change				
Market-implied reading	1.02	1.21	1.04	1.30
Spending-equivalent reading	1.01	1.04	1.05	1.11
Difference D	−0.01	−0.17	0.01	−0.19

Notes: Averages of the 12-month changes in the three assets relative to r^* over the months of each episode, read with the market weights of each definition (points of the two-year yield) and with the FRB/US weights normalized for the same definition (points on the common two-year-yield scale); D is the difference. Baseline intervals are in the text and in Table C.3.

The disagreement is similar under every definition: the ratio of the mean absolute difference to the mean absolute market move in 12-month changes is 0.33, 0.36, 0.42 and 0.39. The contrast in points excludes zero through 2019 and without the lower-bound years and includes it under the two-year yield, so that of the four definitions the interval excludes zero for three.

D.2 The Policy-Rate Experiment in FRB/US

A policy shock in the model gives a comparison with the announcement response. I shock the equation of the inertial Taylor rule in the first quarter only, with the rule active afterwards and expectations from the model's VAR, and I scale the shock so that the average simulated funds rate over the next eight quarters rises by one point. FRB/US has no two-year yield, so that average is my proxy for the unit of policy news. Per unit of the proxy, the 10-year yield rises 1.33 times, equity wealth falls 4.01 times and the dollar rises 0.55 times as much as in the announcement response. GDP four quarters later falls by −0.82 percent, a funds-rate equivalent of 1.77, against 0.95 for the usual announcement response. Priced along their paths, the 10-year yield, equities and the dollar account for 20, 43 and 13 percent of the GDP effect of the model's policy shock, against 32, 25 and 44 percent for the announcement response, and 23 percent is left in the policy rate, spreads and house prices. The model's own policy transmission is heavier in equities and lighter in the dollar than the announcement response, and the difference in the funds-rate equivalent, 1.77 against 0.95, is the residual between the two experiments. The comparison has limits. The funds-rate average is a proxy, and a different proxy changes the unit. The equity fall on impact is large because stock wealth in FRB/US follows expected funds rates through the required return. The exercise is the model's own policy shock, not a reproduction of an announcement. The reading of about one in Section 3.5 is therefore not the statement that announcements move asset prices as a policy shock does in the model; it is a statement about the combined effect on GDP of different mixes.